

Polymetallic veins—a possible bridge between porphyry and intermediate-sulfidation deposits: A case study from Xiongkun ore district, Tibet, China

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ABSTRACT

Understanding the poorly constrained transition between porphyry and intermediate-sulfidation epithermal mineralizations is critical for deciphering the genetic link and fluid evolution in porphyry systems. In this study, we investigate late-stage, post-porphyry polymetallic (Zn-Ag-Au-Cu ± Pb) veins from the No. 1 porphyry Cu-Au deposit in the Xiongkun ore district, Gangdese belt, Tibet, China, which represents “transitional” mineralization bridging porphyry and intermediate-sulfidation epithermal systems. Here we combine detailed petrography, in situ trace element geochemistry of sulfides and quartz, and in situ secondary ion mass spectrometry (SIMS) oxygen isotope analysis of vein quartz to constrain the origin of the polymetallic veins. The veins are paragenetically divided into three stages: the first stage (stage A) comprises pyrite + sericite, the second stage (stage B) is characterized by sphalerite + pyrrhotite + chalcopyrite + galena + quartz, while the third stage (stage C) consists of actinolite + epidote + chlorite + magnetite + quartz. The precipitation of Zn, Ag, Au, Cu, and Pb mainly occurred during stage B. Silver is mainly incorporated in chalcopyrite but also subordinately in acanthite-argentite, pyrrargyrite, or hessite, while electrum inclusions, the only Au-bearing phase, are commonly hosted by chalcopyrite and sphalerite. Mineral geochemical compositions and associated mineral assemblages suggest that the ore-forming fluid displays decreasing temperatures (~317 °C) as well as

decreasing oxygen and sulfur fugacities (fO_2 and fS_2) from stage A to B, thereby triggering the precipitation of sulfides rich in Zn, Ag, Au, Cu, and Pb. The late-stage fluid underwent a further temperature and fS_2 decrease and a pH increase during the subsequent stage C. In situ trace element compositions of vein ore minerals and $\delta^{18}O$ values (−4.2‰ to +4.8‰) of the ore-forming fluid, calculated by in situ analysis of quartz from the polymetallic veins, suggest that the decrease in temperature, $\log fO_2$, and $\log fS_2$ and the increase in pH of the ore-forming fluid are due to the mixing of metal-rich magmatic fluid separated from the magma and relatively reduced alkaline groundwater. This mixing process is also thought to be responsible for the precipitation of sulfides rich in Cu, Au, and Ag along fractures and filling open space. The metal-rich magmatic fluid may have lost some metals (e.g., Cu) during the deep-seated interaction with alkaline groundwater prior to reaching an epithermal environment, thereby prohibiting the formation of high-grade Cu intermediate-sulfidation epithermal deposits. This study provides new insight into the transitional evolution of porphyry and polymetallic mineralizations, thereby improving our understanding of the genetic relationship of porphyry and epithermal mineralizations in porphyry systems and providing a guide for mineral exploration.

1. INTRODUCTION

Polymetallic Zn-Pb-Cu-Ag-Au vein mineralization in porphyry copper systems has a close genetic relationship with porphyry and epithermal deposits and historically has contributed significantly to world production of these commodities (e.g., Reyes, 1991; Baumgartner

et al., 2008; Catchpole et al., 2011, 2015a; Zhai et al., 2019; Crespo et al., 2020; Tomlinson et al., 2021). These veins occur in the periphery of, adjacent to, or superimposed on mineralized porphyry intrusions and are commonly structurally controlled (Baumgartner et al., 2008; Bendezú and Fontboté, 2009; Mottram et al., 2024). In consideration of their temporal and spatial distributions in the mineralized porphyry systems, several studies suggested that these veins should be classified as “transitional,” reflecting the evolution processes at the porphyry to epithermal transition (e.g., Pudack et al., 2009; LeFort et al., 2011; Maydagán et al., 2015; Catchpole et al., 2015b).

Fluid inclusion data and H-O isotopic compositions of quartz suggest that polymetallic veins form under subepithermal to epithermal conditions between 350 °C to 150 °C from fluids with low to moderate salinities (1–28 wt% NaCl equivalent) (Pudack et al., 2009; LeFort et al., 2011; Catchpole et al., 2011; Reed et al., 2013; Rottier et al., 2018). However, the origin of the low- to moderate-salinity fluids and metal precipitation remains controversial, such that some studies favored a late-stage, single-phase magmatic-derived fluid from which the metals precipitated during continuous cooling until reaching the shallow epithermal part of the hydrothermal system (e.g., Rusk et al., 2008; Pudack et al., 2009; Catchpole et al., 2011, 2015b), whereas other studies concluded that the metal precipitation occurred when rising deep hypersaline metal-rich magmatic fluids mixed with low-salinity meteoric water or groundwater (e.g., Camprubí et al., 2006; Baumgartner et al., 2008; Wilkinson et al., 2013; Maydagán et al., 2015; Rottier et al., 2021). In addition, the occurrence of metal precipitation in these transitional veins suggests that the late-stage fluids lost some of their metal endowment before reaching the

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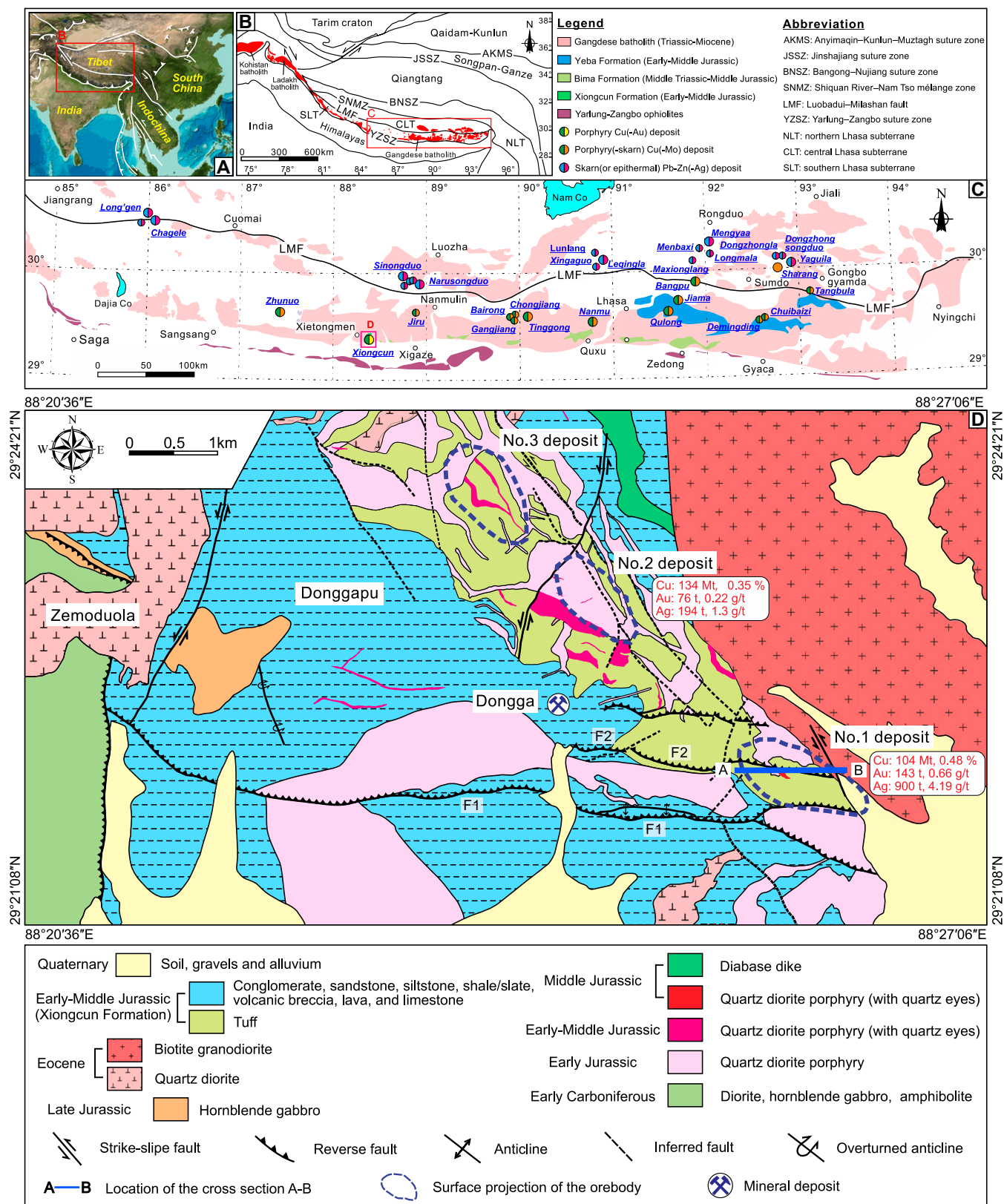


Figure 1. (A, B) Tectonic framework of the Tibetan Plateau (after Zhu et al., 2013). (C) Geological map of the Gangdese porphyry copper belt showing spatial distribution of magmatism and porphyry-related deposits (modified after Hou et al., 2015; Deng et al., 2021). (D) Geologic map of the Xiongcu ore district (modified after Oliver, 2006), showing locations of the No. 1, No. 2, and No. 3 porphyry Cu-Au deposits, Dongga epithermal Au deposit, Zemoduola skarn Cu prospect, and Donggapu porphyry Cu-Au prospect.

epithermal environment, thereby prohibiting the formation of high-grade epithermal deposits at shallower levels (e.g., LeFort et al., 2011). The genetic link between post-porphyry polymetallic veins and porphyry and epithermal mineralizations, however, is still enigmatic.

The Gangdese porphyry copper belt in Tibet, one of the most important Cu-Mo-Au-Ag-Pb-Zn metallogenic belts in China, hosts a number of porphyry Cu(-Au), porphyry(-skarn) Cu(-Mo), and skarn (or epithermal) Pb-Zn-Ag deposits (Zheng et al., 2015; Hou et al., 2015; Yang and Cooke, 2019). The No. 1 deposit (also known as Xietongmen deposit; Tafti et al., 2009) in the Xionggun ore district in the southern Lhasa terrane is a giant porphyry Cu-Au deposit associated with the Neo-Tethyan subduction in the Gangdese porphyry copper belt (Figs. 1 and 2; Lang et al., 2014; Tang et al., 2015; Wang et al., 2017). This deposit comprises polymetallic mineralization overprinting earlier and deeper porphyry-style mineralization in the Gangdese porphyry copper belt. Up to now, previous studies of this deposit mainly focused on the porphyry-type mineralization, including deposit geology, geochronology, petrogenesis, and tectonic setting (Tafti et al., 2009, 2014; Lang et al., 2014; Tang et al., 2015), and fluid evolution and ore genesis (Xu et al., 2009; Tang et al., 2015; Lang et al., 2020). Few fluid inclusion and S-Pb isotope studies have been carried out on the late-stage polymetallic mineralization (Xu et al., 2009; Tang et al., 2015), and, thus, ore-formation processes and the genetic relationships between

the late polymetallic and porphyry-type mineralizations are poorly constrained.

To resolve the above issue, we conducted field mapping, core logging, and detailed sampling of post-porphyry polymetallic veins, which were petrographically and texturally studied. We then collected in situ trace element geochemistry of vein ore minerals, and in situ oxygen isotopes of vein quartz were measured to constrain the origin of the polymetallic veins. This study provides insight into the transitional evolution of the porphyry and polymetallic mineralizations, improving our understanding of the genetic relationship of different mineralizations in porphyry systems, and further provides a guide for mineral exploration.

2. REGIONAL GEOLOGY

From north to south, the Himalayan-Tibetan orogen is composed of the Qaidam-Kunlun, Songpan-Ganze, Qiangtang, Lhasa, and Himalayas terranes (Figs. 1A and 1B; Yin and Harrison, 2000). The Lhasa terrane, bounded by the Bangong-Nujiang suture zone in the north and the Yarlung-Zangbo suture zone in the south, comprises northern, central, and southern subterrane, separated by the Shiquan River–Nam Tso mélange zone and the Luobadui–Milashan fault, respectively (Fig. 1B; Zhu et al., 2011). The northward subduction of the Neo-Tethyan Ocean beneath the Lhasa terrane resulted in the formation of the Middle Triassic–Cretaceous Gangdese batholith and associated arc volcanic rocks

(Fig. 1C; Ji et al., 2012; Kapp and DeCelles, 2019; Deng et al., 2021). The initial collision of India with Asia, reflected by the formation of the Yarlung-Zangbo suture zone, was followed by the breakoff of the Neo-Tethyan oceanic slab, generating the extensive Paleocene–Eocene Linzizong volcanic rocks and coeval intrusions (Zhu et al., 2019; Deng et al., 2024, 2025). The post-collisional magmatism is reflected by ultra-potassic volcanic rocks (25–8 Ma), potassic volcanic rocks (21–10 Ma), and adakitic intrusions (38–10 Ma) (Yang et al., 2016; Liu et al., 2017; Lu et al., 2020, 2023).

The Gangdese porphyry copper belt is situated along the southern margin of the Lhasa terrane and comprises three mineralization episodes (Fig. 1C): (1) Jurassic (180–160 Ma) porphyry Cu-Au mineralization related to the Neo-Tethyan subduction developed in the Xionggun area (Lang et al., 2014; Tafti et al., 2014; Wang et al., 2024); (2) Paleocene–Eocene (65–40 Ma) porphyry-skarn and epithermal mineralization, such as the Sharang porphyry Mo, the Mengya'a skarn Pb-Zn, and the Sinongduo epithermal Pb-Zn-Ag deposits, associated with the Indian-Asian collision along the southern margin of the central Lhasa subterrane (Zhao et al., 2014; Yang et al., 2020); and (3) Miocene (20–10 Ma) Indian-Asian post-collision-related porphyry-skarn Cu(-Mo) deposits, including the Qulong, Jiama, Chongjiang, Tinggong, and Zhunuo Cu(-Mo) deposits (Zheng et al., 2016; Li et al., 2017; Sun et al., 2021, 2024; Lin et al., 2024).

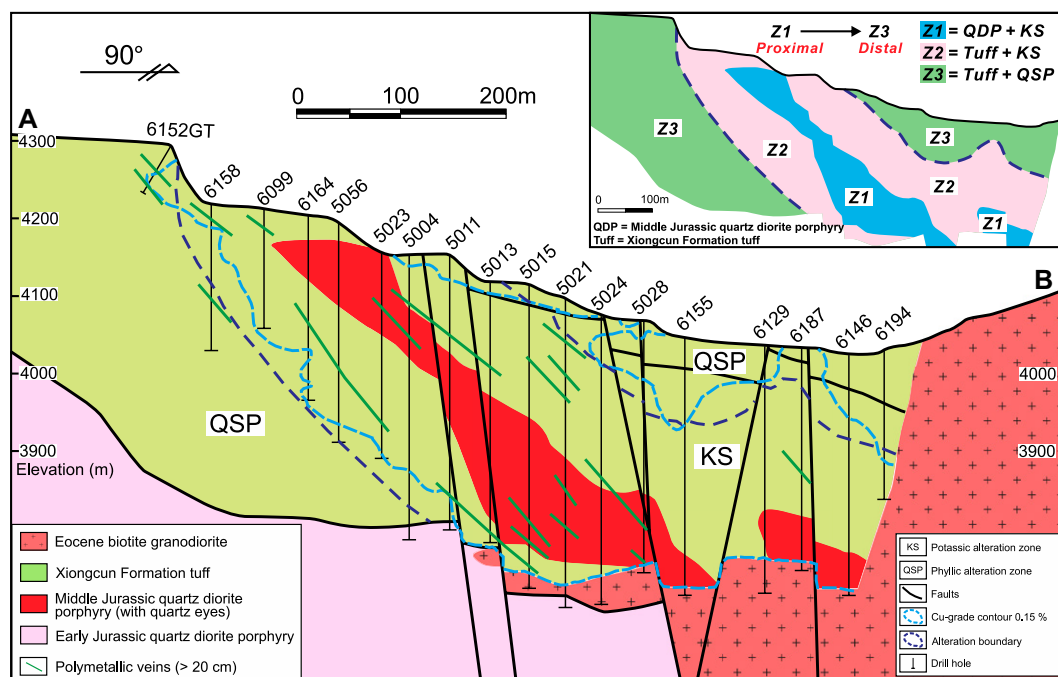


Figure 2. Cross section A–B (see Fig. 1D) of the No. 1 deposit in the Xionggun ore district showing distribution of polymetallic veins. Z1 to Z3 zones (proximal to distal) delineate spatial distribution of polymetallic veins, which is based on alteration and rock types of the No. 1 deposit (see Table 2).

TABLE 1. MAJOR CHARACTERISTICS OF PORPHYRY- AND POLYMETALLIC-TYPE MINERALIZATION FROM THE NO.1 DEPOSIT IN THE XIONGCUN ORE DISTRICT

Mineralization type	Vein type	Alteration type	Mineral assemblage	Vein distribution	Vein thickness
Porphyry type	Early quartz-sulfide veins	Potassic (Bt + As + Qtz + Kfs + Mus + Mag)	Py-Ccp ± Po	Porphyry	<10 cm
	Biotite-sulfide veins		Py-Ccp-Po	Porphyry, tuff	<0.3 cm
	Magnetite-sulfide veins		Mag-Py-Ccp	Porphyry	Typically <0.3 cm, up to 2 cm
	Quartz-molybdenite-sulfide veins		Mo ± Py ± Ccp	Porphyry, tuff	Typically <0.1 cm, up to 1 cm
	Quartz-sulfide veins	Phyllic (Qtz + Py + Ser)	Py-Ccp ± Po	Porphyry, tuff	Typically <10 cm, up to 20 cm
	Pyrite veins		Py ± Ccp ± Po	Porphyry, tuff	Typically <2 cm, up to 5 cm
Polymetallic type	Polymetallic veins	Qtz + Act + Ep + Chl + Cal + Mag	Sp-Py-Po-Ccp-Gn	Porphyry, tuff	Typically <10 cm, up to 2 m

Note: Act—actinolite; As—andalusite; Bt—biotite; Cal—calcite; Ccp—chalcopyrite; Chl—chlorite; Ep—epidote; Gn—galena; Kfs—K-feldspar; Mag—magnetite; Mo—molybdenite; Mus—muscovite; Po—pyrrhotite; Py—pyrite; Qtz—quartz; Ser—sericite; Sp—sphalerite.

3. ORE DEPOSIT GEOLOGY

The Xiongcun ore district, situated at the southern margin of the Gangdese porphyry copper belt, ~50 km west of the Xigaze, consists of the No. 1 (Xietongmen), No. 2 (Newtongmen), and No. 3 porphyry Cu-Au deposits, the Dongga epithermal Au deposit, the Donggapu porphyry Cu-Au prospect, and the Zemoduola skarn Cu prospect (Tang et al., 2012; Tafti et al., 2014; Lang et al., 2019a). The Xiongcun ore district is structurally complex, including Early Carboniferous to Eocene rock units (Fig. 1D). The Early–Middle Jurassic Xiongcun Formation (195–165 Ma; Lang et al., 2019b) is widely exposed in this district and comprises conglomerate, sandstone, siltstone, shale and/or slate, volcanic breccia, lava, and limestone. Intrusive rocks in this area include Early Carboniferous hornblende gabbro and diorite (355–344 Ma; Wang et al., 2020a), Early Jurassic quartz diorite porphyry (181–175 Ma; Lang et al., 2014; Tafti et al., 2014), Early–Middle Jurassic quartz diorite porphyry (174 Ma; Lang et al., 2014), Middle Jurassic quartz diorite porphyry (167–161 Ma; Lang et al., 2014) and diabase dike (165 Ma; Lang et al., 2018), and the Eocene quartz diorite pluton (50 Ma; Tafti et al., 2014), as well as biotite granodiorite (48 Ma; Tafti et al., 2014) (Fig. 1D). E-W-trending post-mineralization faults (F1 and F2) and anticlines are the main structures throughout the ore district; other structures include post-mineralization NE-, N-, EW-, and NW-trending fault zones (Fig. 1D).

The No. 1 deposit is located in the southeast corner of the Xiongcun ore district and contains in total 1.04 Mt of Cu, 143.31 t of Au, and 900.43 t of Ag, with an average of 0.48% Cu, 0.66 g/t Au, and 4.19 g/t Ag (Tang et al., 2012). The ore body is 1300 m long and up to 600 m wide with an average vertical thickness of ~250 m (estimated cutoff grade is 0.15% Cu). It is bounded on its hanging wall and footwall by the F1 and F2 faults, respectively, and shows a N-dipping and E-plunging ellipsoidal or fusiform elongated shape at ~280° to 300° (Figs. 1D and 2). The main host rocks of the deposit are the Middle

Jurassic quartz diorite porphyry and the surrounding Xiongcun Formation tuff (Fig. 2). The quartz diorite porphyry is intensely altered, usually obscuring protolith texture and composition (Tafti et al., 2014). The host intrusion is distinguished from the other porphyry intrusions by round to subhedral quartz eyes, which are up to 1.5 cm in diameter and account for 10% to 15% of the mineral modal content (Lang et al., 2020). The wall rocks immediately adjacent to the host intrusion are fine- to medium-grained tuffs that are commonly highly affected by hydrothermal alteration. Two types of mineralization have been identified: the first event caused the early porphyry Cu-Au mineralization, and the second is recorded by the late polymetallic (Zn-Ag-Au-Cu ± Pb) mineralization (Table 1; Xu et al., 2009; Tafti et al., 2014; Tang et al., 2015).

3.1. Porphyry-Type Mineralization

The hydrothermal alteration of the No. 1 porphyry Cu-Au mineralization is characterized by an early extensive potassic event associated with the Middle Jurassic porphyry intrusive phase (Lang et al., 2014). It is represented by abundant secondary biotite, andalusite, and minor K-feldspar and magnetite and hosts stockwork mineralization consisting of quartz-pyrite-chalcopyrite-biotite ± pyrrhotite veins (early quartz-sulfide veins), biotite-pyrite-chalcopyrite-pyrrhotite veins (biotite-sulfide veins), quartz-pyrite-chalcopyrite ± pyrrhotite veins (quartz-sulfide veins), quartz-molybdenite-sulfide veins, and magnetite-pyrite-chalcopyrite veins (magnetite-sulfide veins) (Lang et al., 2020). Disseminated Cu mineralization associated with the potassic alteration is hosted by the quartz diorite porphyry and surrounding tuff in the central part of the deposit. High-precision Re-Os dating of molybdenite from the quartz-molybdenite-sulfide veins suggests that the porphyry Cu-Au mineralization of the No. 1 deposit was formed at ca. 161.5 ± 2.7 Ma (Lang et al., 2014). The potassic alteration is spatially associated with the high-grade Cu-Au mineralization at the center of the deposit, over-

printed by the poorly mineralized phyllic alteration zone, comprising quartz, sericite, and pyrite that varies, respectively, from pervasive to disseminated from the proximal to the distal zones of the deposit (Tang et al., 2015).

3.2. Polymetallic Mineralization

Based on crosscutting relationships, the polymetallic (Zn-Ag-Au-Cu ± Pb) mineralization is the youngest style of mineralization, overprinting the Middle Jurassic quartz diorite porphyry intrusions and crosscutting the porphyry-stage veins and the potassic and phyllic alteration zones (Figs. 2 and 3A–3D; Tang et al., 2015; Lang et al., 2020). The shallow-dipping (37°–59°) polymetallic veins are concentrated in the center of the deposit and mainly display NE sheeted orientations (Fig. 2). They exceed 600 m in vertical extent, ranging from 4300 m elevation at the highest points of the district to 3700 m elevation at the lowest mining levels. The veins are roughly parallel to the NE-trending Middle Jurassic quartz diorite porphyry (Fig. 2) and were interpreted to have initially formed along steeply dipping fractures preceding the main F1 and F2 thrust faults, which along with smaller faults modified and displaced the No. 1 deposit (cf. Tafti et al., 2014). The polymetallic mineralization ranges from narrow fracture infills to veins >2 m in thickness and can be planar, sinuous, or sheared, filled with massive sulfide and/or gangue minerals (Fig. 3).

4. ANALYTICAL METHODS

Thirteen samples, representative of the polymetallic mineralization, were collected from the drill cores (Fig. S1 in the Supplemental Material¹). Over 150 polished thin sections were examined in reflected and transmitted light to further study of the mineralogy. Details of sam-

¹Supplemental Material. Tables S1–S7 and Figures S1–S8. Please visit <https://doi.org/10.1130/GSAB.S.30251335> to access the supplemental material; contact editing@geosociety.org with any questions.

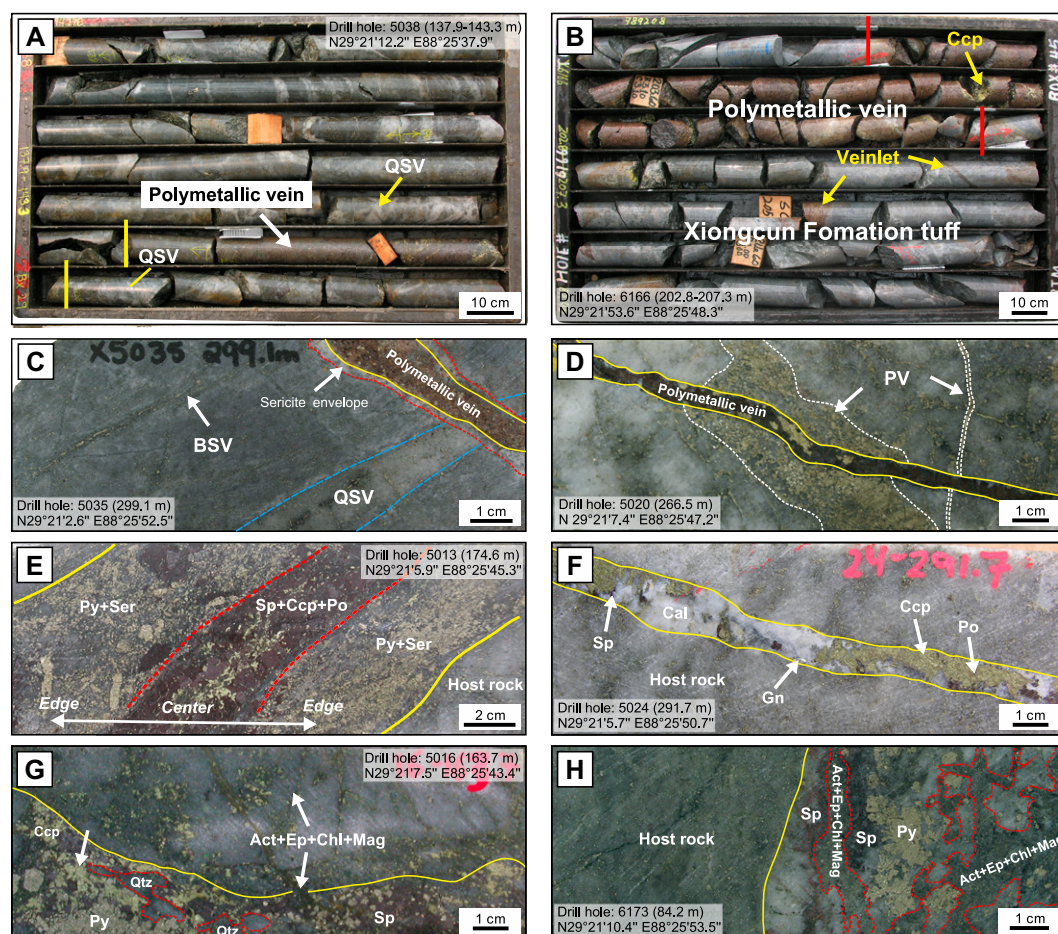


Figure 3. Photographs showing occurrence and texture of polymetallic veins typically observed in the No. 1 deposit. (A) Stockwork of porphyry-stage quartz-sulfide veins (QSV) crosscut by polymetallic veins. The short vertical yellow lines indicate the boundaries between the polymetallic veins and the host rocks. (B) Polymetallic veins cutting early potassic alteration zone within Xiongcu Formation tuff, and smaller veinlets (<5 cm), which are common in adjacent wall rock; both are parallel and perpendicular to main vein. The short vertical red lines indicate the boundaries between the polymetallic veins and the host rocks. (C) Early porphyry-stage biotite-sulfide veins (BSV) and quartz-sulfide veins (QSV) crosscut by polymetallic vein with 3-mm-wide sericite alteration halo. (D) Polymetallic vein with sphalerite and chalcopyrite, crosscutting early porphyry-stage pyrite veins (PV). (E) Polymetallic vein with pyrite-dominated rim domains and abundant sphalerite, chal-

copyrite, and pyrrhotite toward center. (F) Interval of biotite-altered tuff crosscut by polymetallic vein with calcite fillings in intervals between chalcopyrite and sphalerite, pyrrhotite, and galena. (G) Actinolite-epidote-chlorite veinlets crosscutting early-stage pyrite-sericite and sphalerite-chalcopyrite-pyrrhotite-galena veins. (H) Actinolite-epidote-chlorite vein crosscutting sphalerite-chalcopyrite-pyrrhotite-galena vein. Act—actinolite; Cal—calcite; Ccp—chalcopyrite; Chl—chlorite; Ep—epidote; Gn—galena; Mag—magnetite; Po—pyrrhotite; Py—pyrite; Qtz—quartz; Ser—sericite; Sp—sphalerite.

ple locations and analytical methods are summarized in Table S1.

Sulfides were investigated for the presence of micrometer- to nanometer-sized and Au- and Ag-bearing mineral inclusions by using a combination of conventional and field-emission scanning electron microscopy (SEM) techniques. The SEM study was carried out at Wuhan SampleSolution Analytical Technology Co., Ltd., Wuhan, China, using a JEOL IT300HR scanning electron microscope equipped with secondary electron, backscattered electron (BSE), and energy dispersive X-ray spectrometry (EDS) detectors. The analyses were carried out with an accelerating voltage of 15–20 kV and an emission current of ~80 μ A, a takeoff angle ~35°, a 4 μ m to 5 μ m beam diameter, and a working distance of ~10 mm. Semiquantitative EDS spot analyses were used to perform a preliminary chemical assessment of mineral phases and inclusions.

Scanning electron microscopy cathodoluminescence (SEM-CL) images of hydrothermal quartz were obtained to clarify the paragenetic context for the in situ oxygen isotope study. SEM-CL images were acquired at Tuoyan Analytical Technology Co., Ltd., Guangzhou, China, using a TESCAN MIRA3 field-emission scanning electron microprobe (FE-SEM) equipped with a panchromatic cathodoluminescence (CL) detector. SEM-CL images were acquired under an acceleration voltage of 10 kV and a beam current of 15 nA.

Detailed mineralogy and textures were mapped on carbon-coated thin sections using a TESCAN Integrated Mineral Analyzer (TIMA) system at the Nanjing Hongchuang Geological Exploration Technology Service Co., Ltd., Nanjing, China. The TIMA comprises a Mira-3 scanning electron microscope with four EDS (EDAX Element 30) detectors. Samples were scanned

using the TIMA liberation analysis module, and the liberation data including BSE and EDS data were collected with the dot-mapping mode. Individual particles and boundaries between different phases were determined based on the BSE image. Mineral identification and mapping are based on the matching between mineral definition files and EDS data. The volume proportion of all phases were automatically calculated. An accelerating voltage of 25 kV, a probe current of 9 nA, and a working distance of 15 mm were used. Pixel spacing and dot spacing were set as 3 μ m and 9 μ m, respectively.

Sulfide composition was determined by electron probe microanalysis (EPMA) and laser ablation–inductively coupled plasma–mass spectrometry (LA-ICP-MS). EPMA was carried out at Wuhan SampleSolution Analytical Technology Co. using a JEOL JXA-8230 electron microprobe, which was operated at a beam current of

TABLE 2. DETAILED CHARACTERISTICS OF POLYMETALLIC VEINS IN DIFFERENT ZONES

Zone	Host rocks	Alteration zone	Width (cm)	Strike	Dip	Mineral assemblage	Grade
Z1	QDP	KS	0.5–160	NE	38°–52°	Py > Po > Sp > Ccp > Qtz > Act > Chl > Ep > Gn	Cu: 0.15%–2.1% (mean 0.63%) Au: 0.22–4.8 g/t (mean 1.0 g/t) Ag: 1.1–29 g/t (mean 8.5 g/t)
Z2	Tuff	KS	0.7–210	NE	37°–59°	Sp > Py > Po > Qtz > Ccp > Act > Chl > Ep > Gn > Cal > (Ser)	Cu: 0.16%–1.2% (mean 0.58%) Au: 0.11–2.3 g/t (mean 0.91 g/t) Ag: 0.60–26 g/t (mean 9.0 g/t)
Z3	Tuff	QSP	0.3–240	NE	45°–56°	Sp > Qtz > Py > Po > Gn > Cal > Act > Ep > Chl > (Ccp)	Cu: 0.01%–0.84% (mean 0.09%) Au: 0.03–2.0 g/t (mean 0.28 g/t) Ag: 0.20–52 g/t (mean 4.5 g/t)

Note: "Alteration zone" represents early alteration zones that the polymetallic veins overprinted. QDP—Middle Jurassic quartz diorite porphyry; KS—potassic alteration zone; QSP—phylic alteration zone. Mineral abbreviations: Act—actinolite; Cal—calcite; Ccp—chalcopyrite; Chl—chlorite; Ep—epidote; Gn—galena; Po—pyrrhotite; Py—pyrite; Qtz—quartz; Ser—sericite; Sp—sphalerite. Parentheses indicate that the mineral occurs only in trace amounts or is observed in some samples but absent in others.

20 nA, an accelerating voltage of 15 kV, and a beam diameter of 1 μm . Standard specimens used for calibration were Bi_3Se_3 (Bi, Se), Sb_2Te_3 (Sb, Te), metal manganese (Mn), Cu_5FeS_4 (Cu, Fe), ZnS (Zn), FeAsS (As, S), PbS (Pb), metal silver (Ag), and AgAu (Au). Counting times of 20 s on peak and 10 s on the background of both sides of the peak were used for all elements. LA-ICP-MS analyses of sulfides were conducted using a 193 nm ArF excimer laser ablation system (NWR 193) coupled with a Thermo Fisher iCAP RQ at Tuoyan Analytical Technology Co. Helium was used as the carrier gas. Argon was used as the make-up gas and mixed with the carrier gas via a T-connector before entering the inductively coupled plasma-mass spectrometer (ICP-MS). The spot size and frequency of the laser were set to 24 μm and 5 Hz, respectively. The analysis time was 80 s, including 40 s background acquisition and 40 s of sample data acquisition. The ICP-MS was set to time-resolved method and automatically triggered by the laser. The following elements were detected: ^{34}S , ^{45}Sc , ^{49}Ti , ^{51}V , ^{53}Cr , ^{55}Mn , ^{57}Fe , ^{59}Co , ^{60}Ni , ^{65}Cu , ^{66}Zn , ^{69}Ga , ^{74}Ge , ^{75}As , ^{77}Se , ^{97}Mo , ^{105}Pd , ^{107}Ag , ^{111}Cd , ^{115}In , ^{118}Sn , ^{121}Sb , ^{125}Te , ^{133}Cs , ^{182}W , ^{197}Au , ^{202}Hg , ^{205}Tl , ^{208}Pb , and ^{209}Bi . Data reduction was conducted using software Iolite version 3.73 (Hellstrom et al., 2008), with Fe concentration values (EPMA) being used as an internal standard and standards MASS-1, NIST SRM 610, and GSE-2G as external standards.

The in situ trace element of quartz by femtosecond laser ablation-inductively coupled plasma-mass spectrometry (fLA-ICP-MS) was undertaken at the Key Laboratory of Mineralogy and Metallogeny, Guangzhou Institute of Geochemistry, Chinese Academy of Sciences, Guangzhou, China. The J200 Tandem XQ (Applied Spectra) femtosecond laser ablation injection system and the Thermo Scientific iCAP RQ ICP-MS were used for analysis. Compared to the 193 nm laser, the femtosecond laser offers higher spatial resolution and minimizes elemental fractionation and matrix effects (Hergenröder et al., 2006). During analysis, He was used as

the carrier gas and Ar as the auxiliary gas, and the single point analysis time was 105 s, including 20 s of blank background collection, 45 s of continuous erosion collection, and 40 s of cleaning the sampling system. The laser beam diameter and frequency were set to 40 μm and 6 Hz, respectively. NIST SRM 610 was used as the external standard, and Si was used as the internal standard. The international standards used during the measurements are NIST SRM 610, NIST SRM 612, and BHVO-2G, and a group of standard samples was analyzed every eight sample points in order to check the reproducibility and accuracy. The data results were processed with ICPMSDataCal version 12.2 (Liu et al., 2010) by an internal standard method.

The target quartz grains on three selected thin sections of the polymetallic mineralization were cut off then embedded into epoxy resin, polished, and coated with gold to ensure the sample was smooth and conductive. The in situ oxygen isotopes of selected spots ($n = 111$) in nine quartz grains from three quartz samples were analyzed with a CAMECA IMS1280 SIMS at the State Key Laboratory of Isotope Geochemistry, Guangzhou Institute of Geochemistry, Chinese Academy of Sciences, Guangzhou, China. The Cs^+ primary ion beam was accelerated at 10 kV, with an intensity of ~ 2 nA, and rastered over a 20 μm area. The spot size as an ellipse was $\sim 10 \times 20 \mu\text{m}$ in diameter. Oxygen isotopes were measured using multicollection mode on two off-axis Faraday cups, with the intensity of ^{16}O typically 10^9 corrected counts per second (cps) during this study. The instrumental mass fractionation factor (IMF) was corrected using the NBS-28 standard with a $\delta^{18}\text{O}$ value relative to Vienna standard mean ocean water, VSMOW) of 9.5‰ (Matsuhisa, 1974). Measured $^{18}\text{O}/^{16}\text{O}$ ratios were normalized to the VSMOW composition ($^{18}\text{O}/^{16}\text{O} = 0.0020052$) and then corrected for the IMF. The uncertainty on individual analysis was usually better than 0.3‰–0.4‰ (2 standard deviations [SD]). The studied samples were run during the same period as a previous study (Li et al., 2017), and the long-term drift during

the course of study was monitored by analyzing the Qinghu quartz standard between samples. The Qinghu quartz standard yields $\delta^{18}\text{O}$ values between $8.34\text{‰} \pm 0.23\text{‰}$ and $8.92\text{‰} \pm 0.37\text{‰}$, with a mean of $8.64\text{‰} \pm 0.28\text{‰}$ (2SD, $n = 41$; Li et al., 2018).

5. RESULTS

5.1. Ore Mineralogy and Paragenesis

The polymetallic veins were investigated for macroscopic- and microscopic-scale variations in structure, texture, and mineral assemblage in three zones, Z1, Z2, and Z3, of the No. 1 deposit, which correspond to their proximal and distal location in the deposit, respectively (Table 2; Fig. 2).

The polymetallic veins typically occur in groups or clusters, and the mineral assemblages occur either discretely within a vein or as individual veins with clear crosscutting relationships (Figs. 3E, 3G, and 3H). Veinlets, with widths of < 5 cm, commonly form a stockwork and are uniformly distributed on both sides of tabular veins (> 50 cm in width) (Figs. 3A and 3B), which usually have millimeter-thick alteration halos of sericite (Fig. 3C). Contacts between the polymetallic veins and the wall rocks are usually sharp (Figs. 3A and 3B). Sphalerite and chalcopyrite generally occur along the median line of these veins; however, in some veins, large pyrite grains may distribute along the outer edges, getting finer grained toward the median line (Fig. 3E). Interstitial gangue minerals in the veins are quartz and calcite with minor amounts of sericite (Figs. 3F and 3G). Most ore minerals are overprinted or intersected by a late-stage hydrothermal assemblage of actinolite, epidote, chlorite, and magnetite (Figs. 3G and 3H). In addition, the type of mineral assemblage of the polymetallic veins largely depends on the distance to the center of the deposit. The Z1 and Z2 veins dominantly contain pyrite, pyrrhotite, sphalerite, and chalcopyrite, whereas the Z3 veins are mainly composed of sphalerite, quartz,

and pyrite. The calcite abundance in the veins increases sharply with increasing distance from the center of the deposit.

Many microscopic-scale characteristics are common in the studied polymetallic veins regardless of the distance from the center of the deposit. Based on macroscopic- and microscopic-scale observations (Figs. 4 and 5), the polymetallic mineralization is divided into three stages, i.e., early pyrite + sericite (stage A) followed by sphalerite + pyrrhotite + chalcopyrite + galena + quartz (stage B) and late actinolite + epidote + chlorite + magnetite + quartz (stage C) (Fig. 6). The textural and mineralogical features of each stage are as follows. In stage A, euhedral-subhedral pyrite crystals (typically 200–500 μm , up to 2 mm in size) form between 10 vol% and 20 vol% of the sulfides. In stage B, sphalerite, coexisting with pyrrhotite, chalcopyrite, and galena, principally occurs as subhedral to anhedral granular aggregates (Fig. 4F–4H). Pyrrhotite and chalcopyrite occur as disseminated fine-grained exsolution inclusions in sphalerite (Fig. 4I), while the presence of several Au- and Ag-bearing mineral inclusions in chalcopyrite and sphalerite was revealed by SEM imaging and analyses (Figs. 5C–5F; Fig. S2). Electrum (AuAg) commonly forms tiny anhedral inclusions (1–5 μm in diameter) in chalcopyrite (Fig. 5C). Anhedral, small grains (generally 1–2 μm in diameter) of Ag sulfide (Ag_2S , acanthite-argentite), Ag sulfosalt (Ag_3SbS_3 , pyrrargyrite), and Ag tellurides (Ag_2Te , hessite), all of which are characteristic for intermediate-sulfidation epithermal deposits (cf. Wang et al., 2019), occur along the contact between chalcopyrite and sphalerite (Figs. 5D–5F). Furthermore, pyrite is replaced either by chalcopyrite, pyrrhotite, and sphalerite along inter- and intragrain fractures (Figs. 4A–4C) or by sphalerite or quartz forming perforated fine-grained pyrite with patches of sphalerite and quartz (Figs. 4D, 4E, 5A, and 5B). In stage C, the late hydrothermal minerals (actinolite, epidote, chlorite, and magnetite) replace all sulfides along small cracks and surround the previously formed euhedral pyrites (Figs. 4 and 5A).

5.2. Chemical Characteristics of the Polymetallic Veins

5.2.1. Whole-Rock Geochemistry of Polymetallic Vein-Bearing Samples

Representative whole-rock drill core samples that contain sulfide-rich polymetallic veins were collected by Oliver (2006), and their whole-rock geochemical data and descriptions are given in Table S2. The samples have higher Ag, Au, Cu, Zn, Sb, and Pb concentrations than the surrounding low-grade quartz diorite porphyry rocks and

the tuff in the potassic and phyllic alteration zones (Figs. S3 and S4), suggesting that their enriched Ag, Au, Cu, Zn, Sb and Pb contents are likely controlled by the polymetallic mineralization rather than the early porphyry-type mineralization and that these whole-rock samples approximately represent the sulfide-rich polymetallic vein material. Gold concentrations strongly correlate with Ag and Cu within the Z1 and Z2 zones (Fig. 7), which suggests that electrum, associated with chalcopyrite, is the main Au- and Ag-bearing mineral phase, consistent with the SEM imaging. In addition, Cu and Au significantly decrease with increasing distance from the center of the deposit in the veins, indicating that deposition of Cu, Au, and Ag occurred in the Z1 and Z2 zones. On the contrary, within Z3, Cu and Ag as well as Ag and Au concentrations show weak positive correlations of the veins and the high-grade Ag mineralization (up to 50 g/t) that is spatially linked with the veins (Fig. 7). This suggests that Ag is likely present as individual Ag-bearing minerals rather than as electrum or as a trace element incorporated in the vein chalcopyrite. In addition, relative to zones Z1 and Z2, the sharpest overall Sb increase (up to 232 ppm), accompanied by a high Ag content (Table S2), occurs in Z3. Although pyrrargyrite is dominant in the Z3 zone of the veins, hessite and/or argentite could still be major Ag-bearing minerals, given that some veins have relatively high Ag and low Sb concentrations as revealed by SEM imaging.

However, relative to the surrounding wall rocks, the most distinct increase of all enriched ore-forming elements (Cu, Au, and Ag) occurs in the polymetallic veins, which is accompanied by a shift in metal ratios in the three zones (Table S2): high Ag/Au ratios (up to 73; reflecting the low slope of the fitted curve in Fig. 7C) for the veins in the Z3 zone are characteristic for an intermediate-sulfidation epithermal environment (Einaudi et al., 2003), while in contrast, the veins in the Z1 and Z2 zones have relatively low Ag/Au ratios (up to 32). The Cu/Ag ratios also show strong variations in the Z1, Z2, and Z3 zones: polymetallic veins in the Z1 (average 821) and Z2 (average 942) zones have higher Cu/Ag ratios than in the Z3 zone (average 285) (Fig. 7B). Consequently, the high-grade Cu-Au-Ag mineralization in the No. 1 deposit is interpreted to be the result of the late polymetallic mineralization overprint.

5.2.2. Chemical Characteristics of Polymetallic Vein Sulfides

The EPMA and LA-ICP-MS data are reported in Tables S3 and S4. The LA-ICP-MS sulfide analyses can be affected by the presence of (sub)microscale inclusions containing various

elements (cf. Román et al., 2019). As a result, all LA-ICP-MS spectra were inspected for the presence of spikes indicating mineral inclusions, and the LA-ICP-MS dataset for each sulfide was filtered. However, it should be noted that the scanning capability of LA-ICP-MS is unable to distinguish between homogeneously distributed nano-inclusions and lattice-bound elements because both types of trace element speciation display flat and uniform patterns (e.g., Wohlgemuth-Ueberwasser et al., 2015; Gregory et al., 2015). Compositional variations of pyrite, pyrrhotite, chalcopyrite, and sphalerite from the Z1, Z2, and Z3 zones are described in the next four subsections. A summary of trace element concentrations for each sulfide is presented in Figure 8.

Pyrite. The measured pyrite grains contain 44.92–46.82 wt% Fe, with sulfur contents ranging from 52.57 wt% to 54.34 wt% (EPMA analyses; Table S3). Pyrite has a low trace element content: Co, Ni, As, and Se were detected in >90% (above the limit of quantification) of the analyzed spots, Mn, Te, Pb, and Bi in >30% of the analyzed spots, and Au, Ag, and Tl locally (LA-ICP-MS analyses; Table S4). Arsenic (As) in most of the samples shows flat and smooth patterns in the LA-ICP-MS transient signals and mean concentrations largely >10 ppm. In contrast, Au, when detected, shows a spiky pattern and no correlation with Ag (Fig. S5A). Therefore, Au probably occurs as minor electrum inclusions, thereby implying that As does not have a coupled incorporation with Au and Ag, which is different from typical epithermal Au-Ag deposits (e.g., Deditius et al., 2014). This is because As substitutes for S^{2-} in the pyrite lattice, leading to lattice expansion and thereby providing space for the large Au ion (Reich et al., 2005). Furthermore, high Au concentrations also occur in As-free pyrite in the form of telluride inclusions in some high-sulfidation epithermal deposits (Cook et al., 2009); however, the telluride inclusion-bearing pyrites analyzed here lack detectable concentrations of Au by LA-ICP-MS analysis (Table S4). Cu shows no clear correlation with other analyzed elements (Table S4), which favors the hypothesis that Cu is structurally bound, replacing Fe in the octahedral sites of the pyrite structure (Einaudi, 1968). For example, as much as 8 wt% Cu can be structurally bound in hydrothermal pyrite (Pačevski et al., 2008); additionally, Cu may also be present as microscale chalcopyrite inclusions (Deditius et al., 2011). Zn displayed flat and uniform patterns during LA-ICP-MS analysis and, thus, may be incorporated in independent nanoscale inclusions in pyrite (cf. Deditius et al., 2011; Gregory et al., 2015). Besides, the obtained LA-ICP-MS transient signals, showing spiky pat-

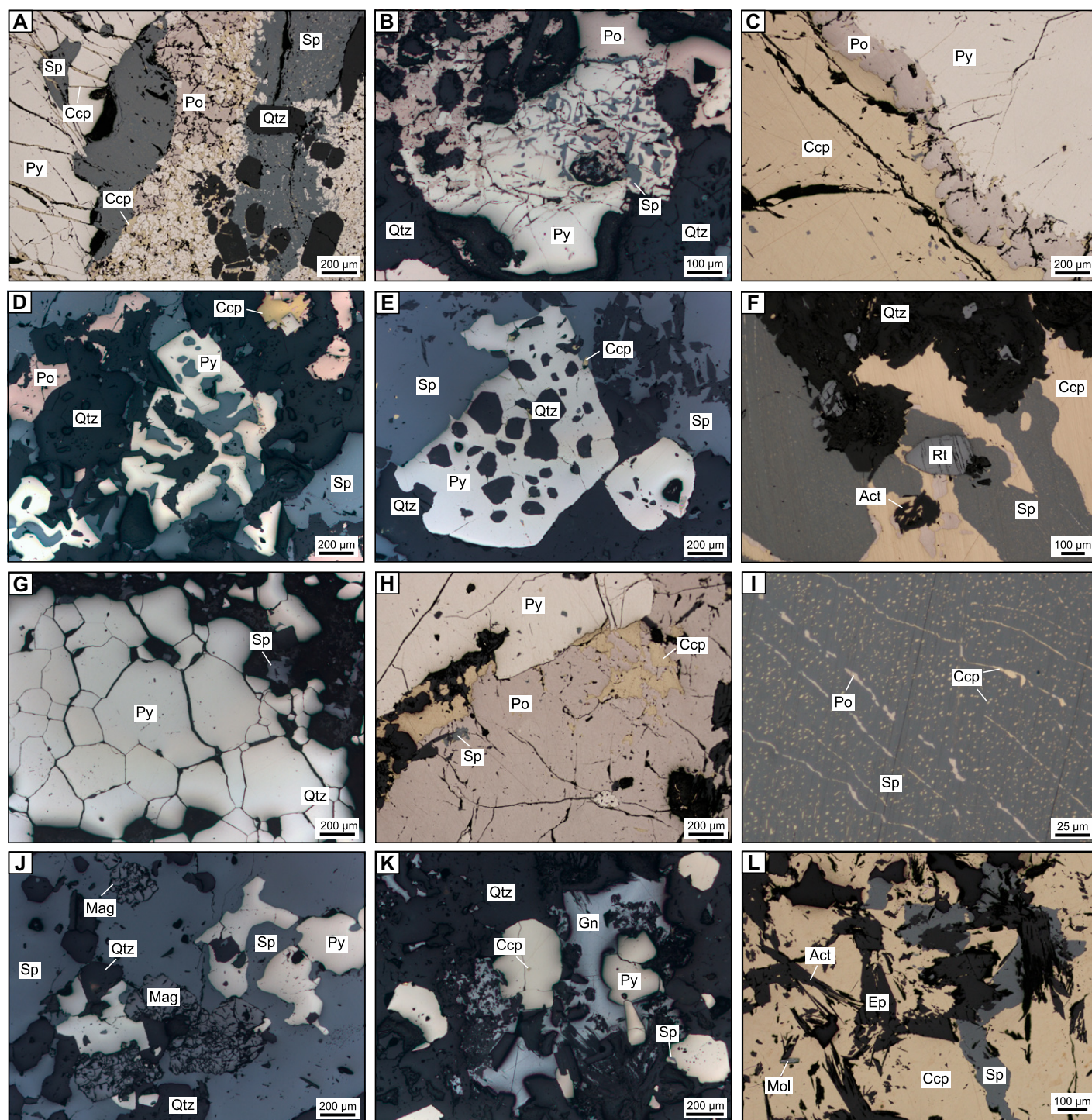


Figure 4. Reflected-light photomicrographs illustrating mineralogy and texture of sulfide ores from polymetallic veins in the No. 1 deposit. (A–C) Pyrite replaced by late sphalerite, chalcopyrite, and pyrrhotite, which are mainly concentrated along pyrite boundaries. (D, E) Sphalerite and quartz replacing euhedral pyrite, forming poriferous texture. (F) Rutile, quartz, and actinolite assemblage, replacing chalcopyrite and sphalerite. (G) Massive euhedral pyrite distributed along edge of polymetallic veins that were not replaced by other sulfides. (H) Aggregates of sphalerite, chalcopyrite, and pyrrhotite are separated by smooth contacts, lacking corrosion or replacement. (I) Sphalerite enclosing chalcopyrite and pyrrhotite inclusions as ordered chains of ellipsoidal beads. (J) Magnetite and quartz replacing large aggregates of sphalerite and pyrite. (K) Anhydrous galena replaced by late quartz. (L) Late actinolite and epidote overprinting chalcopyrite, sphalerite, and molybdenite assemblage. Act—actinolite; Ccp—chalcopyrite; Ep—epidote; Gn—galena; Mag—magnetite; Mol—molybdenite; Po—pyrrhotite; Py—pyrite; Qtz—quartz; Rt—rutile; Sp—sphalerite.

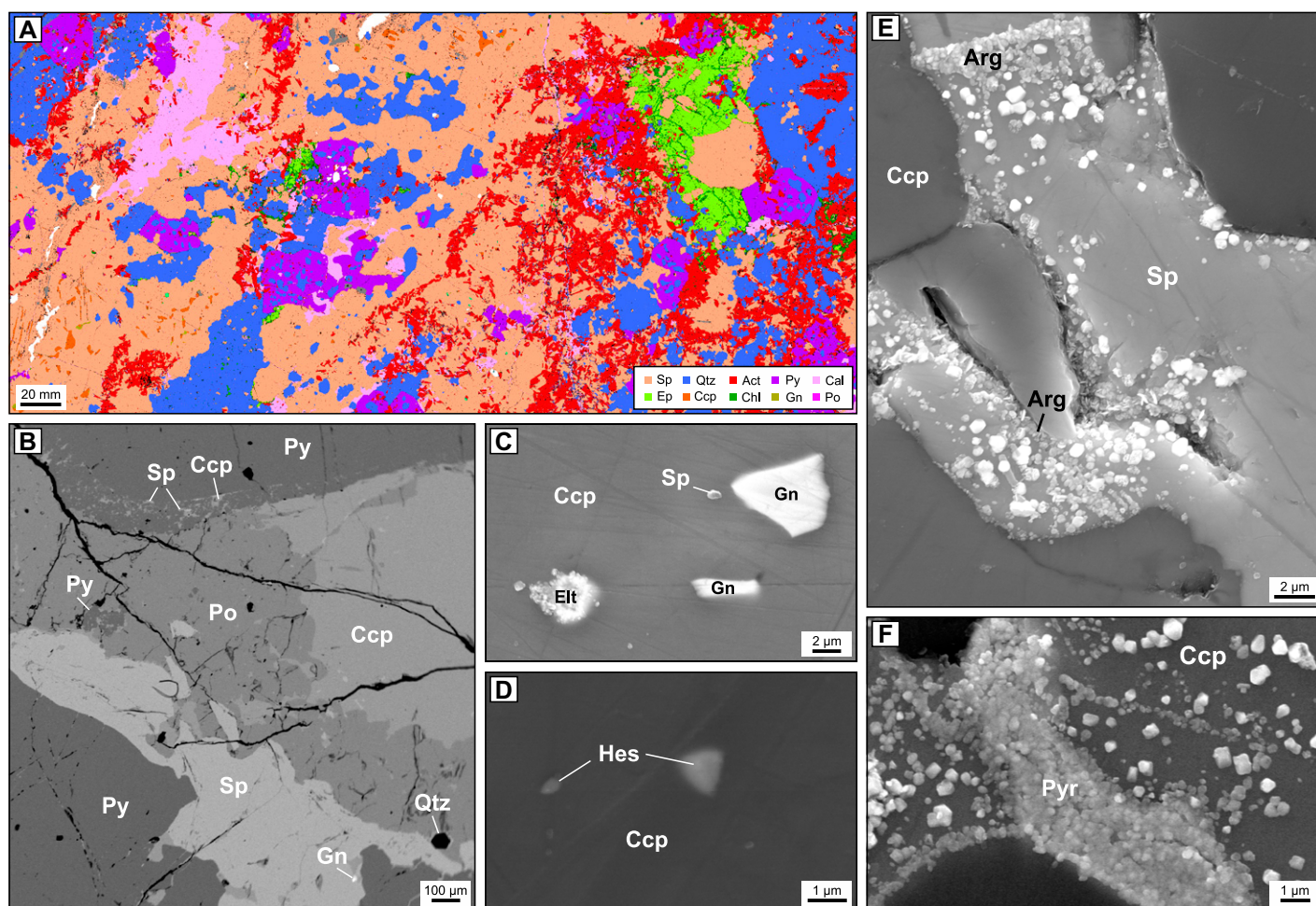


Figure 5. Microscale features of sulfides in polymetallic veins in the No. 1 deposit. (A) Mineral mapping of polymetallic vein samples by TESCAN Integrated Mineral Analyzer (TIMA) showing mineral assemblages and their replacement relationship, illustrating sphalerite, chalcopyrite, pyrrhotite, and galena replacing early pyrite, which is subsequently replaced by late actinolite, chlorite, epidote, and calcite. (B) Backscattered electron image showing pyrite grain replacement by sphalerite, pyrrhotite, and chalcopyrite with remnant patches and perforation. (C, D) Scanning electron microscopy (SEM) images reflecting electron and hessite inclusions in chalcopyrite. (E, F) SEM images of argentite and pyrrargyrite inclusions distributed in rim domains of sphalerite and chalcopyrite crystals. SEM–energy dispersive X-ray spectrometry analysis results are provided in Figure S2 (see text footnote 1). Act—actinolite; Arg—argentite; Cal—calcite; Ccp—chalcopyrite; Chl—chlorite; Elt—electron; Ep—epidote; Gn—galena; Hes—hessite; Po—pyrrhotite; Py—pyrite; Pyr—pyrrargyrite; Qtz—quartz; Sp—sphalerite.

terns for Pb, suggest the presence of small galena inclusions (Fig. S5A). In summary, Ni, As, Se, Ag, and Te are typically more enriched in pyrite of the distal zones (Z2 and Z3), whereas Co is typically more enriched in pyrite of the proximal zone (Z1) (Fig. 8A).

Pyrrhotite. Pyrrhotite has major element concentrations ranging from 58.73 wt% to 60.47 wt% Fe and 38.85 wt% to 40.40 wt% S based on EPMA (Table S3). Pyrrhotite grains usually have low trace element concentrations, commonly below the detection limit. Mn, Ni, Se, Pb, and Bi are the only elements detected in >90% of the analyses, and Co is detected in >50% of the analyses (Table S4). The obtained LA-ICP-MS transient signals, showing almost

flat patterns, and the trace element correlations suggest that, except for Fe, Ni, and Se (Fig. S5B), the detected trace element concentrations correspond most probably to small inclusions of galena and rare Bi-Ag-Pb sulfosalt (members of the lillianite-homologous series; Fuertes-Fuente et al., 2011). The pyrrhotite in the polymetallic veins shows that from Z1 to Z3, Se, Ag, and Bi concentrations distally decrease, whereas Co and Ni increase (Fig. 8B).

Chalcopyrite. EPMA analysis of the major elements in chalcopyrite from Z1 to Z3 range from 34.53 wt% to 35.61 wt% S, 33.64 wt% to 34.51 wt% Cu, and 30.11 wt% to 30.82 wt% Fe (Table S3). LA-ICP-MS analyses of chalcopyrite revealed that Mn, Zn, Se, Pd, Ag, Cd,

In, Pb, and Bi are above the detection limit in all analyses and Ga, Sb, Te, and Tl in >50% of the analyzed spots. Ge, Au, and Hg are rarely above the detection limit (Table S4). Most analyzed elements have relatively smooth patterns. Zn and Pb show some spiky signals (Fig. S5C) that do not correlate with those of the other elements, suggesting the presence of sphalerite and galena solid micro-inclusions. Au mainly displays spiky patterns and weakly correlates with Ag, indicating that Au mostly occurs as electrum inclusions. This is in accordance with observations that hydrothermal chalcopyrite is not a preferential host for Au with maximum concentrations usually less than a few parts per million regardless of deposit type (George

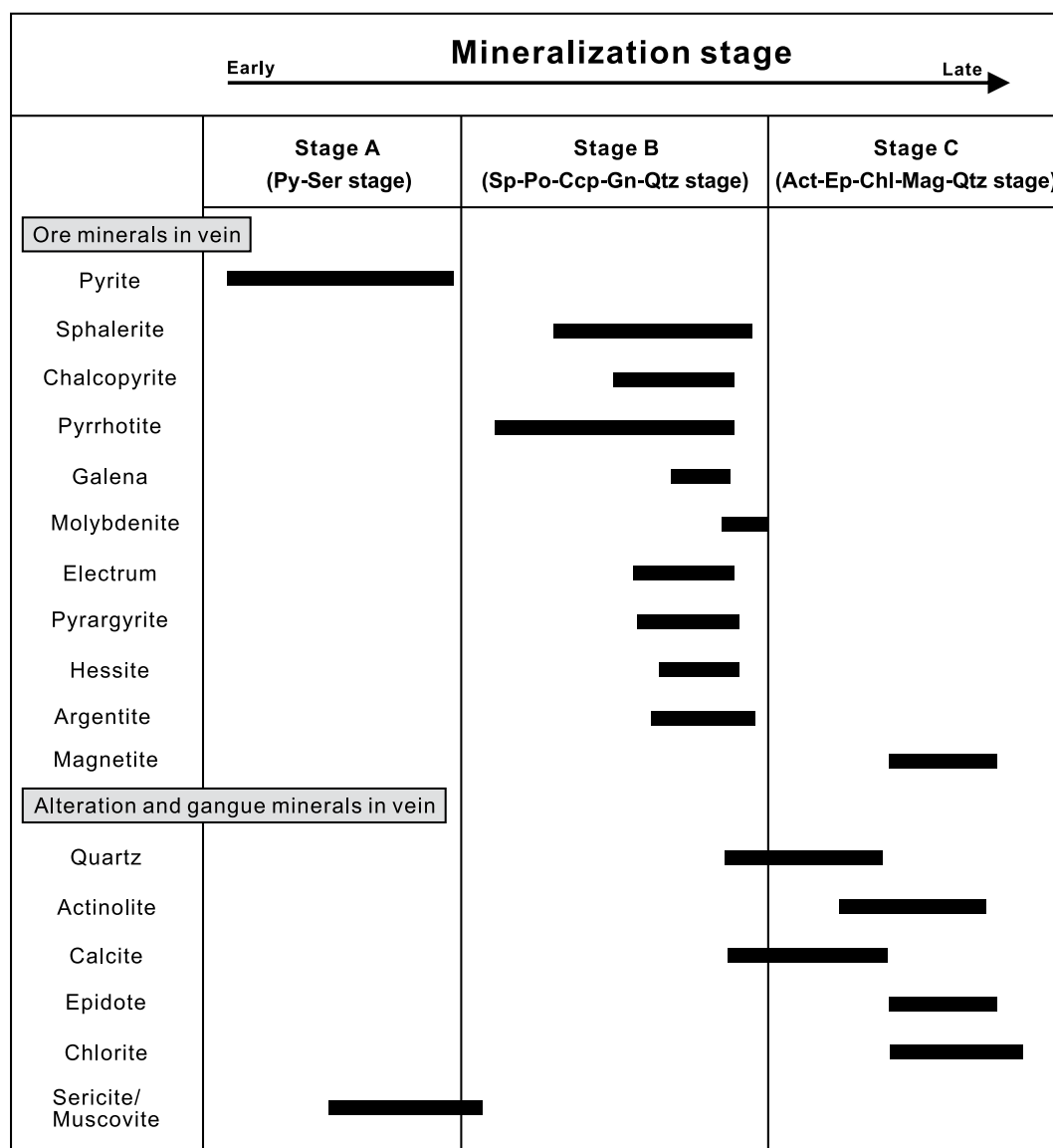


Figure 6. Paragenetic sequence of ore and alteration minerals of the three stages of polymetallic mineralization in the No. 1 deposit. Act—actinolite; Ccp—chalcopyrite; Chl—chlorite; Ep—epidote; Gn—galena; Mag—magnetite; Po—pyrrhotite; Py—pyrite; Qtz—quartz; Ser—sericite; Sp—sphalerite.

et al., 2016). In addition, Zn, Sn, In, and Ag generally show smooth variations in transient signals (Fig. S5C), suggesting that incorporation into the chalcopyrite lattice took place by substitution, because the ionic radii of Zn^{2+} , Sn^{4+} , and In^{3+} in tetrahedral coordination fall within a “window” between the ionic radii of Fe^{3+} and Fe^{2+} , and Ag^+ is the monovalent ion closest in size to Cu^+ in tetrahedral coordination (George et al., 2018). A strong correlation between Cu and Ag is shown by the whole-rock samples, and the chalcopyrite abundance decreases from the Z1 to the Z3 zone of the deposit, along with significantly decreasing Ag concentrations (Fig. 7; Fig. S6), revealing that most Ag in the polymetallic veins is incorporated in chalcopyrite. It is evident that Ga, Se, Ag, Te, Au, Tl, and Bi concentrations

in the chalcopyrite decrease from the proximal toward the distal zones of the deposit, whereas Cd, In, and Sn all increase (Fig. 8C).

Sphalerite. Sphalerite grains have variable major element concentrations ranging from 50.67 wt% to 64.50 wt% Zn, 32.80 wt% to 34.25 wt% S, and 1.98 wt% to 11.19 wt% Fe that were analyzed by EPMA (Table S3). Trace elements Mn, Ga, Se, Ag, Cd, In, Hg, Pb, and Bi were detected in >90% and Ge, Sn, and Tl in >50% of the analyzed spots; V, Mo, and Au were locally detected (above the limit of quantification) (Table S4). Similar trace element enrichments occur in sphalerite from epithermal deposits associated with other porphyry systems (e.g., Maydagán et al., 2013). Mn, Cd, In, Hg, and Sn in most of the samples show flat and smooth patterns. Co, Fe, and Pb present a

spiky pattern lacking a correlation with other elements (Fig. S5D), thereby suggesting that they occur mostly as pyrite and galena solid inclusions. Cu, when detected, shows a spiky pattern and correlates with Fe and Ag (Fig. S5D); it probably occurs as minor inclusions of chalcopyrite.

Principal component analysis of trace element contents in sphalerite reveals distinct correlations between certain elements, which can discriminate between different substitution mechanisms (cf. Koch, 2013). In simpler terms, PCA helps visualize which elements behave similarly or differently in the crystal structure. The contrasting loadings of Hg, Mn, Cd, and Fe suggest that they occupy different sites or substitute for Zn in different ways, which supports our interpretation of distinct substitution mecha-

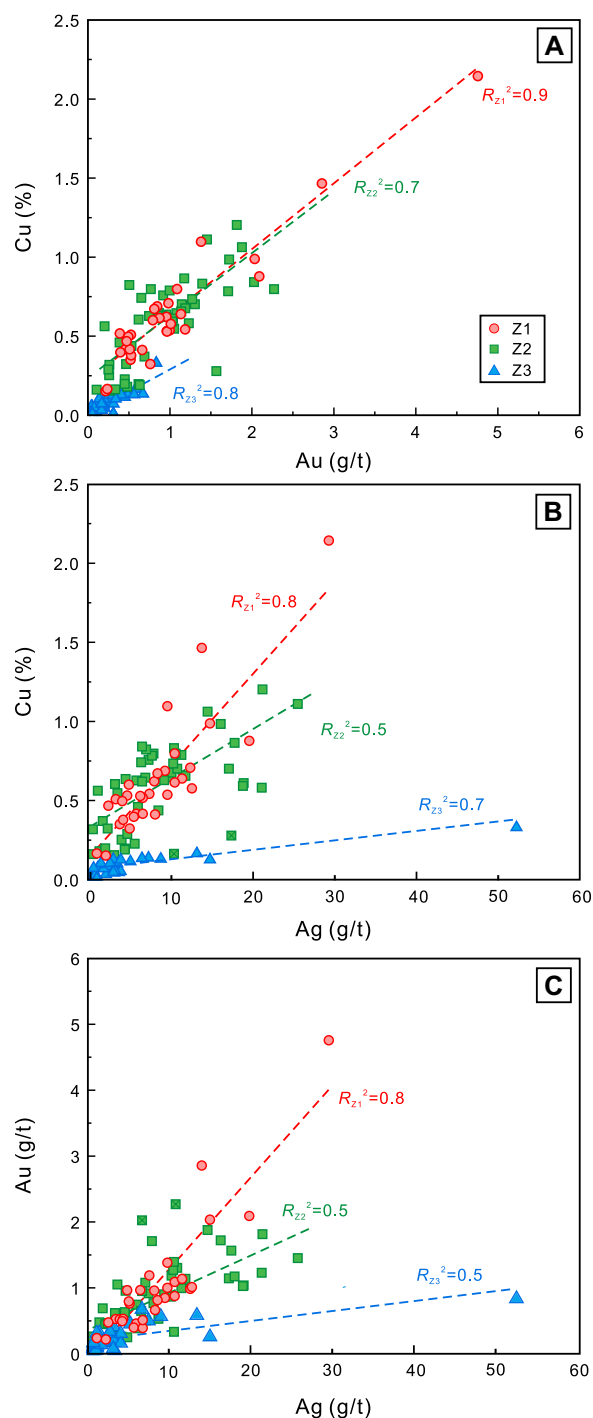


Figure 7. Binary plots of Cu versus Au (A), Cu versus Ag (B), and Au versus Ag (C) concentrations in whole-rock samples of polymetallic vein-filling material and associated wall rocks. Z1, Z2, and Z3 refer to polymetallic veins in Zone 1, Zone 2, and Zone 3, respectively. Crosses in symbols represent discrete data points and are excluded from calculation of trend line. Analytical data are provided in Table S2 (see text footnote 1).

nisms. Usually, Zn^{2+} is replaced by Fe^{2+} , Cd^{2+} , Mn^{2+} , Hg^{2+} , and Sn^{2+} (Seifert and Sandmann, 2006; Murakami and Ishihara, 2013). However, the Hg, Mn, Cd, and Fe contents in the sphalerite studied here display distinct loadings with opposite signs on the principal components (Fig. S7), which is in accordance with the suggested direct substitutions ($\text{Zn}^{2+} \leftrightarrow \text{Hg}^{2+}$, $\text{Zn}^{2+} \leftrightarrow \text{Mn}^{2+}$, $\text{Zn}^{2+} \leftrightarrow \text{Cd}^{2+}$, $\text{Zn}^{2+} \leftrightarrow \text{Fe}^{2+}$). Besides, Ge, Ga, and Cu correlate in sphalerite (Fig. S7), sug-

gesting a substitution mechanism such as $3\text{Zn}^{2+} \leftrightarrow 2\text{Cu}^{+} + \text{Ge}^{4+}$ and $\text{Cu}^{+} + \text{Ga}^{3+} \leftrightarrow 2\text{Zn}^{2+}$, while the correlation between Pb and Sb (Fig. S7) suggests the $4\text{Zn}^{2+} \leftrightarrow \text{Pb}^{2+} + 2\text{Sb}^{3+} + \square$ (vacancy) substitution. Furthermore, the strong correlation among Sn, In, and Ag (Fig. S7) indicates three possible substitutions: $3\text{Zn}^{2+} \leftrightarrow \text{In}^{3+} + \text{Sn}^{3+} + \square$, $3\text{Zn}^{2+} \leftrightarrow \text{In}^{3+} + \text{Sn}^{2+} + \text{Ag}^{+}$, and $4\text{Zn}^{2+} \leftrightarrow \text{In}^{3+} + \text{Sn}^{4+} + \text{Ag}^{+} + \square$. The Se, Pd, Ag, Sn, Hg, Pb, and Bi concentrations of the

sphalerite from the different zones of this deposit display a decrease distally within the deposit, whereas Mn, Ga, and Tl all increase (Fig. 8D).

5.2.3. Trace Element Compositions and Oxygen Isotopes of Quartz

The SEM-CL imaging revealed only one quartz generation in the polymetallic veins (Fig. 9). Typical CL images show quartz as CL-black fine-grained euhedral grain fillings and as inclusions enclosed in sulfides, which corresponds with our hand sample observations (Fig. 9). The quartz grains from the Z1 to Z3 zones show strong variations of Li, B, Al, and Ti, but several important trace elements in quartz (e.g., Ga, Ge, and Sb; Rottier and Casanova, 2021) have low concentrations (Table S6). The $\delta^{18}\text{O}$ values of quartz grains from the Z1 zone range from 9.74‰ to 10.93‰, with a mean of 10.30‰. In contrast, the Z2 and Z3 quartz grains yield $\delta^{18}\text{O}$ values ranging from 9.37‰ to 10.78‰ (mean 10.19‰) and from 9.30‰ to 10.54‰ (mean 9.78‰), respectively. Thus, both the Z2 and Z3 quartz grains are slightly depleted in ^{18}O relative to the quartz grains from the Z1 zone (Fig. 9; Table S7).

6. DISCUSSION

6.1. Physicochemical Conditions of Mineralization

The precipitation of sulfides in magmatic-hydrothermal systems is interpreted to be due to changes of temperature (T), sulfur fugacity (f_{S_2}), oxygen fugacity (f_{O_2}), and pH conditions (Einaudi et al., 2003; Hurtig et al., 2021). Most studies have suggested that the Se content of pyrite increases with increasing fluid temperature (Maslennikov et al., 2009; Revan et al., 2014), while some studies have suggested that Se in pyrite depends on the pH and the redox conditions of the ore-forming fluids (Yamamoto et al., 1984; Huston et al., 1995). However, Keith et al. (2018)—in accordance with the former studies—reported that the Se content in pyrite is mainly a function of the fluid temperature and introduced a new geothermometer using the equation:

$$\text{Se}_{\text{pyrite}} = 5 \times 10^{13} \times T^{-4.82}. \quad (1)$$

The concentrations of Se (ranging from 5 to 480 ppm, generally <100 ppm) in pyrite from the polymetallic veins during stage A are consistent with that in porphyry and epithermal deposits (Huston et al., 1995; Cepedal et al., 2008; Reich et al., 2013), indicating that most of the pyrite may have formed at temperatures >300 °C (mean 324 °C; Table S4), similar to

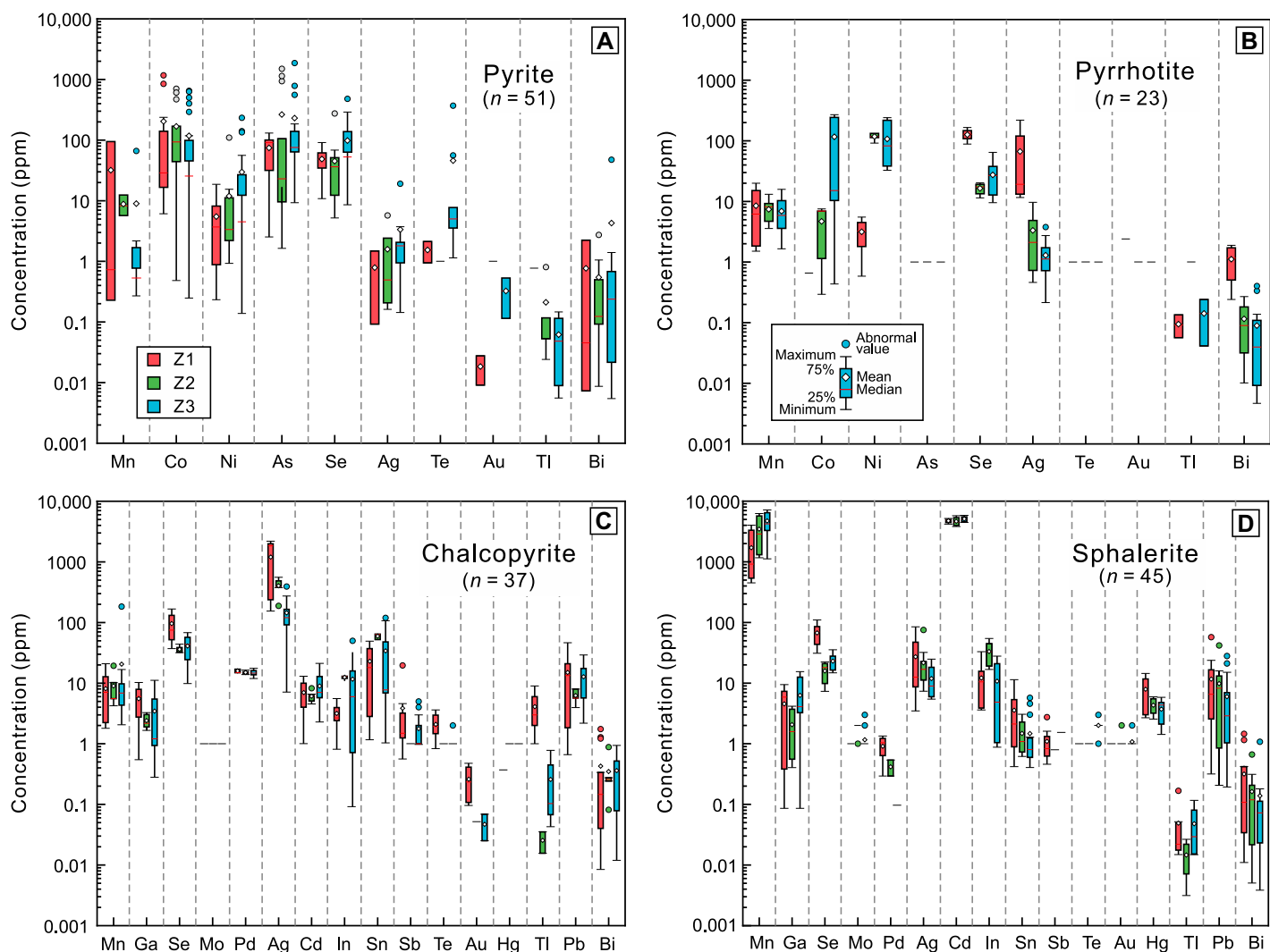


Figure 8. Box-and-whisker plots of trace element compositions in pyrite, pyrrhotite, chalcopyrite, and sphalerite grains from Z1, Z2, and Z3 zones, which are ordered from left to right by increasing distance from center of deposit. All results are available in Table S4 (see text footnote 1).

pyrite of D-type veins in porphyry mineralizations (e.g., Crespo et al., 2020; Wang et al., 2020b). Furthermore, the pyrite from the Z3 zone is enriched in Se compared to that in the Z1 zone (Fig. 8A), suggesting that the fluid temperature for proximal precipitated pyrite significantly exceeds that in the distal zones of the deposit. In addition, the formation temperature of the sphalerite was determined using the GGIMFis geothermometer (Frenzel et al., 2016), which is based on the Ga, Ge, Fe, Mn, and In contents of the sphalerite, following the equations:

$$PC1^* = \left(\frac{C_{Ga}^{0.22} \times C_{Ge}^{0.22}}{C_{Fe}^{0.37} \times C_{Mn}^{0.20} \times C_{In}^{0.11}} \right), \text{ and} \quad (2)$$

$$T(^{\circ}\text{C}) = -(54.4 \pm 7.3) \times PC1^* + (208 \pm 10), \quad (3)$$

where PC1* is the first principal component score derived from the trace element contents of sphalerite, and C represents the concentration of each element (in ppm). Using these equations, we obtained temperatures of $317 \pm 25^{\circ}\text{C}$ (Table S4), indicating that the formation temperature of the stage B sphalerite is similar to that of stage A pyrite (mean 324°C). Further temperature constraints come from the mineral assemblage of pyrite, chalcopyrite, galena, sphalerite, and Ag-bearing sulfosalt, indicating temperatures $<350^{\circ}\text{C}$ and an intermediate-sulfidation environment (e.g., Einaudi et al., 2003; Mercer and Reed, 2013; Wang et al., 2019). Consequently, the Zn-Ag-Au-Cu $\pm$ Pb precipitation of stage B is thought to have occurred at temperatures between 350°C and 300°C , suggesting cooling fluids as the hydrothermal system waned.

In general, the Ti incorporation in quartz is known to be mainly dependent on temperature, and it is generally regarded as a credible geothermometer (e.g., Huang and Audétat, 2012; Rottier et al., 2021). The stage C paragenesis contains rutile (Fig. 4F) that buffers the titanium activity of the fluid at 1 ($a_{\text{TiO}_2} \approx 1$; cf. Rottier et al., 2021). Calculated stage C quartz precipitation temperatures range between 174°C and 321°C (average 248°C ; Table S6) when using the equation of Zhang et al. (2020). Consequently, the minerals in the polymetallic veins are estimated to have most likely formed under low- to medium-temperature conditions (~ 180 – 350°C), with a possible trend of decreasing temperatures from stage A to stage C. A similar temperature range (200 – 300°C) was obtained for polymetallic vein mineraliza-

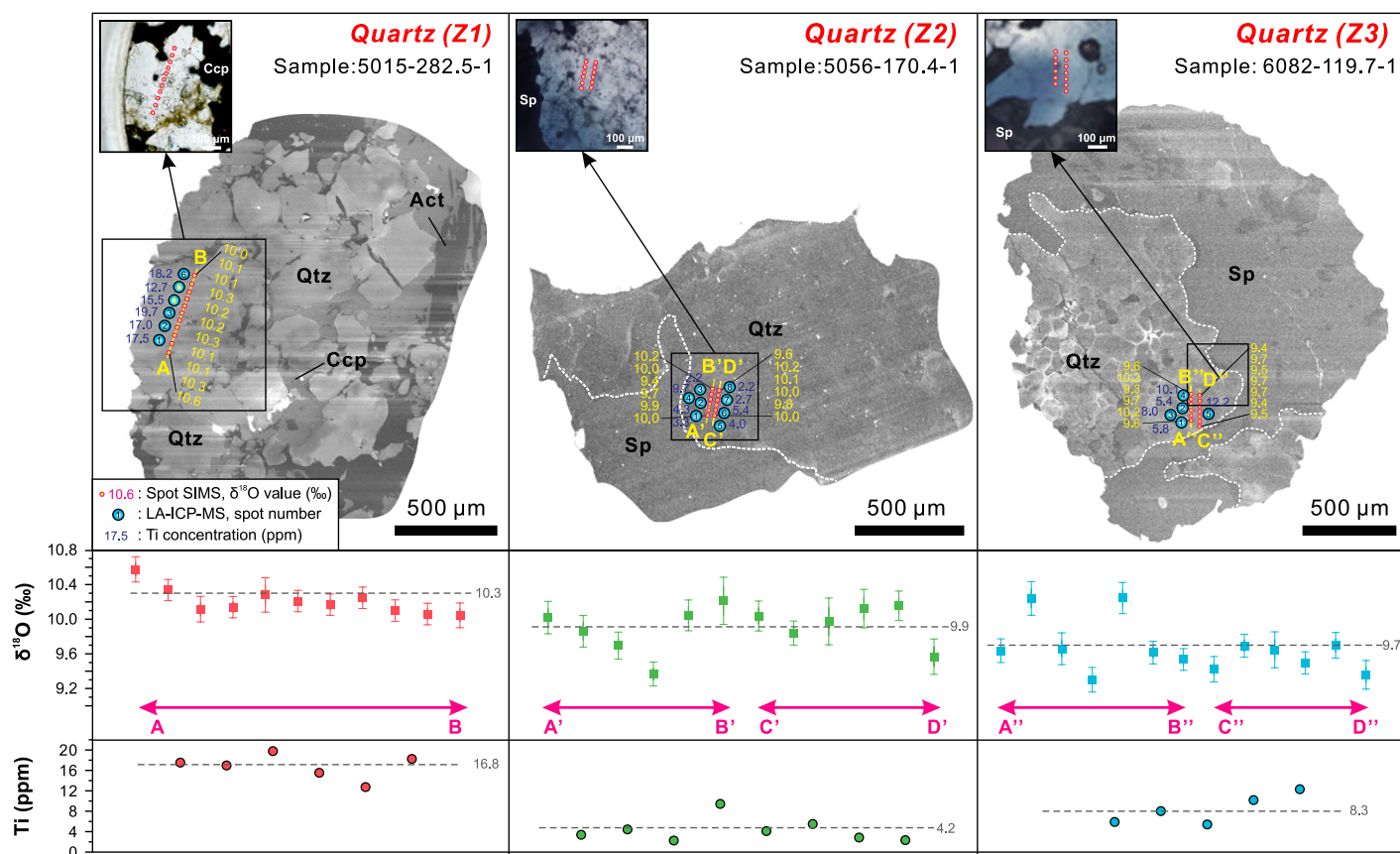


Figure 9. Scanning electron microscopy cathodoluminescence images of quartz grains from polymetallic veins (Z1 to Z3 zones) in the No. 1 deposit, showing positions of test spots and their $\delta^{18}\text{O}$ values and Ti compositions (SIMS—secondary ion mass spectrometry; LA-ICP-MS—laser ablation–inductively coupled plasma–mass spectrometry). The insets at the upper left of each panel show the transmitted-light photomicrographs of the quartz targets with the analytical spots marked. Gray dotted lines on $\delta^{18}\text{O}$ value and Ti composition panels represent averages of $\delta^{18}\text{O}$ values and Ti contents, respectively. Error bars represent two standard errors. Act—actinolite; Ccp—chalcopyrite; Qtz—quartz; Sp—sphalerite.

tion in other porphyry systems using the Ti-in-quartz geothermometer (e.g., Cerro de Pasco deposit, Peru; Rottier et al., 2021).

Changes of the temperature and $f\text{S}_2$ conditions can control the deposition of high-, intermediate-, or low-sulfidation sulfides (Einaudi et al., 2003). Stage A pyrite lies within the intermediate-sulfidation range between the arsenopyrite and pyrrhotite buffer, followed by the deposition of pyrrhotite (stage B), reflecting a progressive decrease of the $f\text{S}_2$, while stage C magnetite falls within the low-sulfidation range between the pyrrhotite and fayalite buffer (Fig. 10A). The changes of mineral assemblages from stage A to stage C suggest an evolution path from intermediate- to low-sulfidation states. The diagram of acidity (pH) and oxygen fugacity ($f\text{O}_2$) constructed for porphyry systems (Fig. 10B) was used to estimate the stability fields of minerals in the polymetallic veins assuming equilibrium conditions. For this purpose, minerals were considered as ideal solid solutions. Furthermore,

the pH range of the Zn-Ag-Au-Cu $\pm$ Pb mineralization (stage B) was thought to range from 4.9 to 5.7 based on the presence of pyrrhotite and sericite in the veins. In this pH range, the $f\text{O}_2$ conditions are below those of pyrrhotite stability ($\log f\text{O}_2$ is < -34.7) and pyrite becomes unstable at 300 °C. These conditions correspond to the Zn-Ag-Au-Cu $\pm$ Pb precipitation stage B of the polymetallic vein assemblage. The oxygen fugacity decreased to $\log f\text{O}_2 = -34.7$ during the fluid evolution from stage A to stage B (see Fig. 10B) that led to a decrease of the $\text{HSO}_4^-/\text{H}_2\text{S}$ ratio (Chouinard et al., 2005) and the formation of the Zn-Ag-Au-Cu $\pm$ Pb mineral paragenesis as part of the main ore stage for the polymetallic mineralization. The occurrence of stage C magnetite yields a pH increase (> 8) (Fig. 10B). Moreover, based on the thermodynamic model proposed by Charles et al. (1988), the presence of epidote and chlorite requires pH values above those of the muscovite stability field in the Fe-Cu-S-O system. Furthermore, the pH value

increase was accompanied by a $f\text{S}_2$ decrease during the fluid evolution from stage B to stage C. In conclusion, the $f\text{S}_2$ - T and $f\text{O}_2$ -pH phase diagrams suggest that the fluid underwent a decrease of the oxidation state and $f\text{S}_2$ from stage A to stage B, which was followed by a progressive T and $f\text{S}_2$ decrease and a pH increase until stage C.

6.2. Source of Ore-Forming Fluids

Oxygen isotopes are a common and effective way to trace the possible origin of ore-forming fluids in mineralized porphyry systems (e.g., Cao et al., 2014; Rottier et al., 2021). Assuming that quartz precipitation took place under isotope equilibrium with the aqueous fluid, we calculated $\delta^{18}\text{O}_{\text{H}_2\text{O}}$ values using the Ti-in-quartz geothermometer and the oxygen isotope fractionation factors in the quartz- H_2O system (Clayton et al., 1972). The estimated oxygen isotopic composition of the ore-forming fluid for the polymetallic mineralization is presented

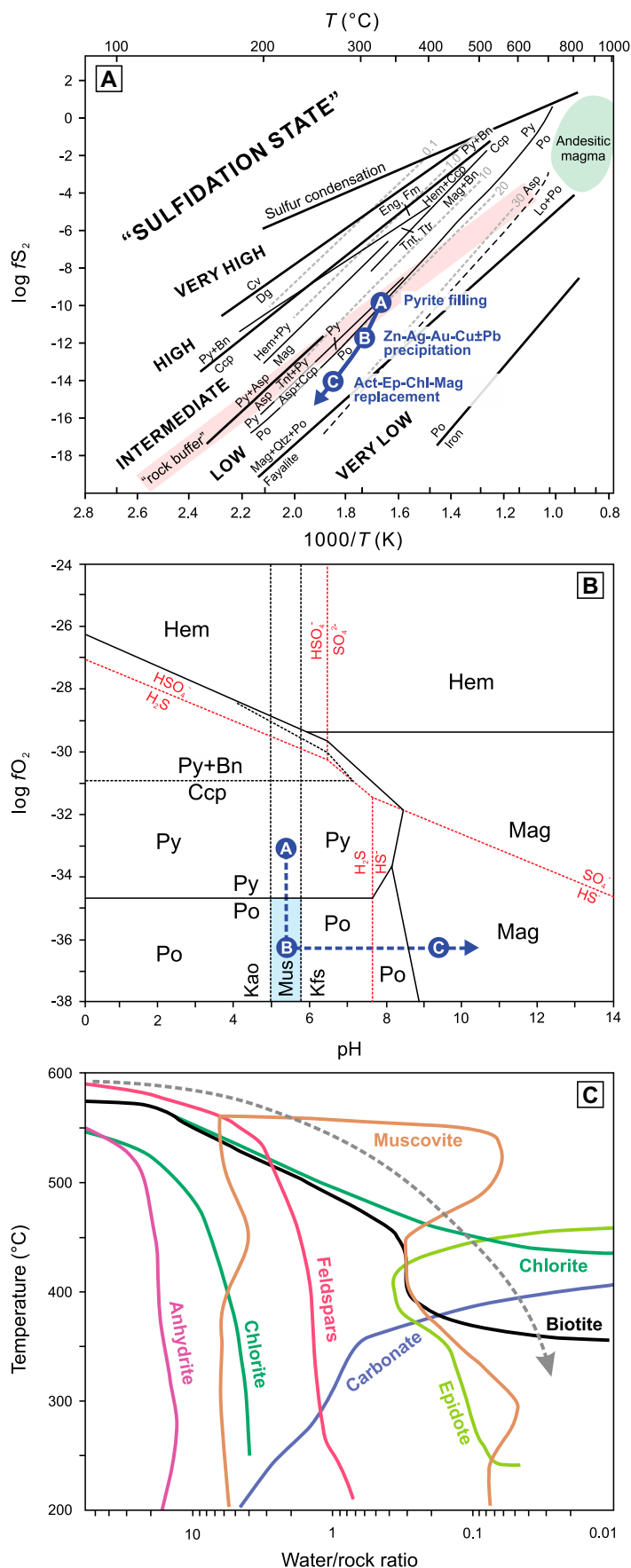


Figure 10. (A) Diagram of $\log fS_2$ versus $1000/T$ (where T is temperature) (modified from Einaudi et al., 2003, and Baumgartner et al., 2008), illustrating evolution of fluid properties during polymetallic mineralization stages A to C. Temperature and fS_2 estimates are based on mineral geothermometry and stability of different sulfide assemblages. (B) Diagram of $\log fO_2$ versus pH, showing stability fields of minerals in the Cu-Fe-S-O system at 300 °C and 20 Mpa. Stability fields were estimated using SUPCRT92 software (Johnson et al., 1992), assuming $\Sigma S = 0.001$ mol/kg and $\Sigma NaCl = 1$ mol/kg. Arrow indicates evolution of ore-forming system. Blue dotted line presents inferred evolving trend in panel B, because estimated temperature is a range (324–248 °C) rather than 300 °C condition. (C) Alteration mineral phase diagram showing temperature conditions and reaction ratio of magmatic fluid to granite (based on data from Butte, Montana, USA) that control presence or absence of alteration minerals (simplified from Reed et al., 2013). Lines on diagram are labeled on the side where the relevant mineral exists, being absent on the opposite side. Gray dotted line represents evolution path of alteration mineral assemblages in porphyry systems. Mineral abbreviations: Act—actinolite; Asp—arsenopyrite; Bn—bornite; Ccp—chalcopyrite; Chl—chlorite; Cv—covellite; Dg—digenite; Eng—enargite; Ep—epidote; Fm—famatinitite; Hem—hematite; Kao—kaolinite; Kfs—K-feldspar; Lo—löllingite; Mag—magnetite; Mus—muscovite; Po—pyrrhotite; Py—pyrite; Qtz—quartz; Tnt—tennantite; Ttr—tetrahedrite.

in Table S7. In contrast, assuming disequilibrium during quartz precipitation, quartz $\delta^{18}O$ values would be higher than under equilibrium conditions (e.g., Clayton and Mayeda, 2009). In general, disequilibrium partitioning of oxygen isotopes in quartz may arise from varying growth rates across the crystallographic faces of a single crystal, known as sector zoning (Tanner et al., 2013), or during periods of rapid growth (Allan and Yardley, 2007). The impact of sector zoning on quartz $\delta^{18}O$ values is not well defined, while some evidence suggests negligible to low influence (<2‰) (Tanner et al., 2013). During periods of fast growth, quartz is prone to more structural defects, resulting in an increase in trace element concentrations (Ihinger and Zink, 2000). However, the quartz crystals studied here are characterized by the absence of significant SEM-CL textures (Fig. 9) and strongly variable

trace element concentrations (Table S6), thereby implying only a minor impact of disequilibrium partitioning of oxygen isotopes in quartz.

The studied samples from the polymetallic mineralization show relatively lower $\delta^{18}\text{O}_{\text{H}_2\text{O}}$ (-4.2‰ to $+4.8\text{‰}$) values than the porphyry-type mineralization in the No. 1 deposit (0.3‰ to 5.2‰ ; Lang et al., 2020), which could be a result of fluid boiling, water-rock interaction, or mixing of magmatic and meteoric water (e.g., Harris et al., 2005; Cao et al., 2014). Fluid boiling can be excluded due to the lack of vein textures characteristic for fluid boiling (e.g., bladed calcite, chalcedony; White and Hedenquist, 1990). This is consistent with a fluid inclusion study revealing the absence of coexisting vapor-rich and liquid-rich inclusions (Xu et al., 2009). Moreover, as a result of boiling, an increase of the liquid $\delta^{18}\text{O}$ values should have occurred (Harris and Golding, 2002) because ^{16}O preferentially fractionates into the vapor phase relative to ^{18}O (Shmulovich et al., 1999). Water-rock interaction should also have generated depleted oxygen isotopes in the residual fluid (cf. Cao et al., 2014). However, the influence of the pervasive hydrothermal alteration prior to polymetallic mineralization on the $\delta^{18}\text{O}$ values of the end-member compositions remains unclear, and, therefore, it is difficult to develop quantitative models to unravel the modification of the $\delta^{18}\text{O}$ values by water-rock interaction. We adopt the water-rock interaction model by Reed et al. (2013) to roughly estimate the water/rock ratios: the progressive sequence of alteration assemblages of the polymetallic vein aureoles emphasizes an evolution path from the stability fields of muscovite and subsequently to chlorite + epidote during decreasing temperatures and water/rock ratios (>0.1 to <0.1 ; Fig. 10C). In addition, some of the polymetallic veins have straight boundaries without significant sericite alteration aureoles (Fig. 3), suggesting a relatively minor influence of water-rock interaction on the oxygen isotopic compositions of the ore-forming fluid.

Therefore, based on the following evidence, we propose that the decrease of the $\delta^{18}\text{O}_{\text{H}_2\text{O}}$ values (i.e., the decrease of polymetallic mineralization relative to porphyry mineralization) is the result of mixing of magmatic and meteoric water: (1) decreasing δD values (decrease to -109) and salinities (decrease to 5.13 wt% NaCl equivalent) of the ore-forming fluid from the porphyry system to the polymetallic mineralization imply a major influx of meteoric water (cf. Xu et al., 2009); (2) the $\delta^{18}\text{O}_{\text{H}_2\text{O}}$ values of the polymetallic mineralization from the proximal to the distal zone of the deposit show a distinct decreasing trend (Fig. 11A), also in accordance with the addition of meteoric water; and (3) the

decrease of Co/Ni ratios of pyrite from the Z1 to the Z3 zone (Fig. 11B) suggests an increasing proportion of meteoric water that interacted with the magmatic fluids with increasing distance to the porphyry center, given that higher Co/Ni ratios tend to be associated with more saline fluids and their precipitates (cf. Keith et al., 2022).

6.3. Sequence of Mineralization

In conjunction with previous studies (Xu et al., 2009; Tafti et al., 2014; Lang et al., 2020), our results suggest a new formation model for the polymetallic veins in the No. 1 deposit (Fig. 12). Middle Jurassic dioritic magma emplacement caused extensional doming in the cover rocks (Xiongcu Formation), while dilatational fault zones provided high-permeability pathways for fluid ascent (cf. Burnham, 1979; Richards, 2011). Considering high volatile concentrations in the upper porphyry zones and the formation of brittle fractures, the fluid is thought to have escaped from the intrusion during pressure decrease, dramatically reducing the solubility of the metal-chloride species and triggering metal deposition, eventually resulting in the formation of early porphyry-type Cu-Au mineral-

ization within the porphyry and surrounding tuff (Fig. 12A; Lang et al., 2020).

As magma solidification advances downward, the entire magmatic-hydrothermal system progressively cools, host and wall rocks undergo brittle deformation (Fournier, 1999), and erosion progressively degrades the paleosurface (Sillitoe, 2010). The steeply dipping fractures commonly align with the regional stress field (Hildreth and Hannah, 1996; Cammans, 2015), and the orientations of the late polymetallic veins in porphyry copper systems reflect this phenomenon (e.g., Catchpole et al., 2015b; Tomlinson et al., 2021). Therefore, the early-formed extensive steep fissure network of the No. 1 deposit is thought to have acted as conduits for the upwelling mineralizing fluids (Fig. 12B), representing high-permeability pathways (Figs. 12C and 12D; cf. Richards, 2011). Pyrite precipitation in the polymetallic veins occurred at the early stage (stage A) and appears to have been formed by filling of void spaces (Figs. 12E and 12H). The sericite alteration, as observed in the margin of the veins, reflects the water-rock interaction within the associated wall rocks.

The subsequent Zn-Cu-Ag-Au $\pm$ Pb (stage B) main mineralization event, characterized

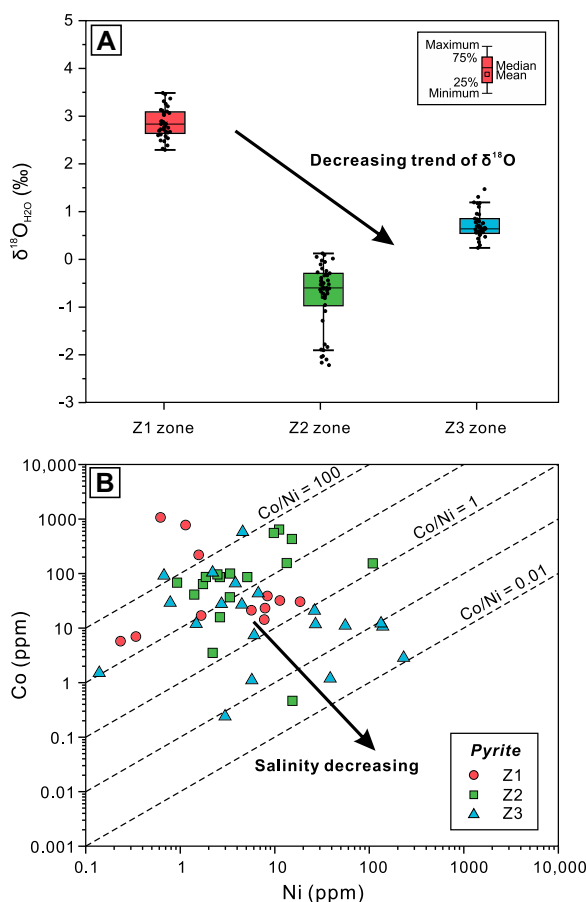
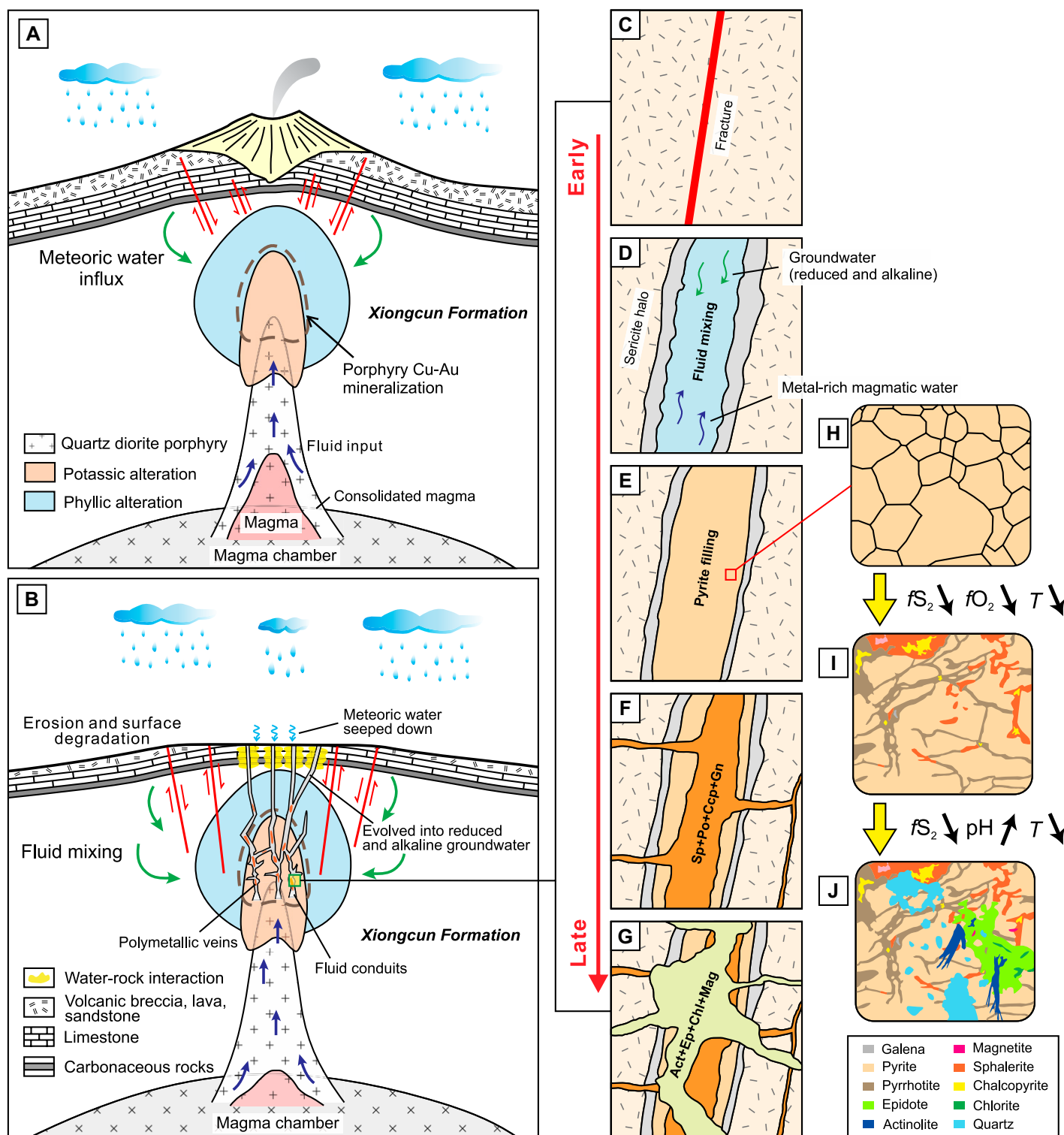


Figure 11. Calculated $\delta^{18}\text{O}_{\text{H}_2\text{O}}$ values (A) and bivariate variation diagram of Co versus Ni in pyrite (B) from Z1, Z2, and Z3 zones of polymetallic mineralization. All results are available in Tables S4 and S7 (see text footnote 1).



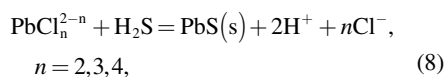
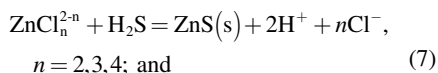
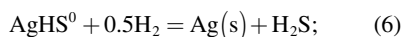
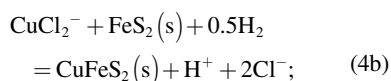
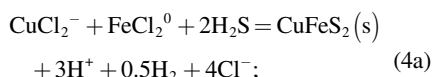
by sphalerite + pyrrhotite + chalcopyrite + galena replacing early pyrite, occurred during a progressive decrease of the temperature, oxidation, and sulfidation state of the ore-forming fluid (Figs. 12F and 12I). Experimental studies demonstrate that gold and silver bisulfide species ($AuHS^0$ and $AgHS^0$) are dominant at weak acid

to near-neutral conditions, whereas gold and silver chloride species ($AuCl_2^-$ and $AgCl_2^-$) are dominant under mildly acidic to acidic conditions at low to moderate temperatures ($<350^\circ C$) (Pal'yanova, 2008; Williams-Jones et al., 2009). In contrast, copper is transported dominantly as chloride complex ($CuCl_2^-$) at a

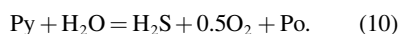
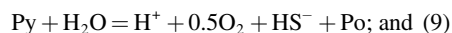
wide range of temperatures ($200\text{--}600^\circ C$) and near-neutral pH conditions (Etschmann et al., 2010; Louvel et al., 2017), whereas copper bisulfide species [$Cu(HS)_2^-$] are generally stable in the low-density vapor phase (Etschmann et al., 2010). Besides, lead and zinc are transported dominantly as chloride complex ($ZnCl_n^{2-n}$ and

Figure 12. Cross sections showing model for polymetallic vein formation and mineralization (see text for more details). (A) Fluid escapes from intrusion during pressure decrease, dramatically reducing solubility of metal-chloride species and triggering metal deposition, eventually resulting in formation of early porphyry-type Cu-Au mineralization within the porphyry and surrounding tuff. (B) As magma solidification advances downward, the entire magmatic-hydrothermal system progressively cools, and rock undergoes brittle fracturing, while erosion progressively degrades paleosurface. Steeply dipping fractures align with regional stress field, and orientations of late polymetallic veins in the porphyry copper system reflect this phenomenon. (C, D) Early-formed extensive steep fissure network acted as conduits for upwelling mineralizing fluids, representing high-permeability pathways. Sericite alteration halo, as observed in margin of veins, reflects intense water-rock interaction with associated wall rocks. (E, H) Pyrite precipitation in polymetallic veins occurred at early stage (stage A) and formed by filling of void spaces. (F, I) Following was progressive decrease of temperature, oxidation, and sulfidation state of ore-forming fluid, the main mineralizing event of Zn-Cu-Ag-Au ± Pb mineralization corresponding to stage B, characterized by sphalerite (Sp) + pyrrhotite (Po) + chalcopyrite (Ccp) + galena (Gn) replacing early pyrite. (G, J) Actinolite (Act) + epidote (Ep) + chlorite (Chl) + magnetite (Mag) assemblage (stage C) replaced early sphalerite + pyrrhotite + chalcopyrite + galena, reflecting decreasing temperature (T) and fS_2 but increasing pH in fluid.

$PbCl_n^{2-n}$; $n = 2, 3$, or 4) at a wide range of temperatures (200–450 °C) under acidic to weakly alkaline conditions (Etschmann et al., 2019; Chai et al., 2024). Therefore, the ore-forming fluid was evaluated to be a medium-temperature (~317 °C), near-neutral (see section 6.1 “Physicochemical Conditions of Mineralization”), and low-density aqueous fluid phase (fluid inclusion data; Xu et al., 2009). As a result, it is thought that Cu, Au, Ag, Zn, and Pb were transported as $CuCl_2^-$, $AuHS^0$, $AgHS^0$, $ZnCl_n^{2-n}$, and $PbCl_n^{2-n}$ complexes, respectively, in the ore-forming fluids. To evaluate possible mechanisms of ore deposition, the following reactions were considered (Kouzmanov and Pokrovski, 2012):



where (s) indicates the solid state. During stage B, chalcopyrite and pyrrhotite (Po) precipitated simultaneously, thereby replacing early pyrite (Py), which can be attributed to reaction 4b and the following reactions (Mishra and Pal, 2008):



These reactions point out that the contemporaneous deposition (stage B) of copper, base

metal sulfides (e.g., Zn, Pb), and gold and silver requires increasing pH and decreasing fO_2 conditions.

On the other hand, the Mn contents in sphalerite are strongly affected by redox conditions (Bernardini et al., 2004; Kelley et al., 2004), with high Mn contents generally reflecting a reduced environment (Kelley et al., 2004; Zhuang et al., 2019). The increasing Mn contents of sphalerite from the Z1 (average = 1974 ppm) to Z2 (average = 3449 ppm) to Z3 (average = 3847 ppm) zones (Fig. 8D) suggest the presence of a more reduced ore-forming fluid in Z3 in comparison to those in Z1 and Z2. In addition, the precipitation of sulfides consumes large amounts of H_2S , while the disproportionation of SO_2 generates abundant H_2SO_4 , both contributing to a more oxidized fluid environment. To counterbalance this and maintain reduced conditions, the addition of a reduced fluid is required. Considering the ore-forming fluid as a mixing product of magmatic and meteoric water, the reduced fluid can be attributed to the involvement of relatively reduced groundwater that evolved from meteoric water. This interpretation is supported by the occurrence of CH_4 -rich fluid inclusions in the polymetallic veins of the No. 1 deposit (Xu et al., 2009), which are thought to have originated from the thermal decomposition of the surrounding carbonaceous shale and slate of the Xiongcu Formation (Lang et al., 2020), further indicating the involvement of reduced groundwater. Likewise, the observation that the calcite abundance increases sharply with increasing distance from the center of the deposit (Table 2) indicates that the involved groundwater was slightly alkaline, because the loss of H^+ due to the neutralization through alkaline groundwater causes an increase in carbonate (liberated from carbonic acid in the fluid), favoring precipitation of calcite (Holland and Malinin, 1979). Although Kokh et al. (2017) suggested that CO_2 -rich fluid pulses from the underlying magma chamber should mix with epithermal aqueous fluid, thereby triggering HCO_3^- decomposition and formation

of carbonates in porphyry systems, fluid inclusions in the quartz of the studied polymetallic veins are characterized by the absence of CO_2 (Xu et al., 2009). Furthermore, the increase in pH values from the proximal to the distal zones also corresponds with the Al contents of quartz in the Z2 (36–749 ppm) and Z3 (49–120 ppm) zones (Table S6), given that some studies proposed that the Al content of quartz negatively correlates with the pH of the mineralizing fluids (Allan and Yardley, 2007; Rusk et al., 2008). It is further noteworthy that due to the deposition of stage B sulfides, the hydrothermal fluid should—according to the reactions above—progressively evolve to a more acidic condition; however, the stage C mineral assemblage indicates a general increase in pH. This pattern of pH evolution is also compatible with a hydrothermal system that is buffered by injection of an alkaline fluid, thereby explaining the low Al contents (27–69 ppm) of quartz in zone Z1, because acidic fluids moved outward and were, thus, gradually neutralized by groundwater, while the Z1 magmatic fluid still maintained a relatively high pH due to magma chamber input. Thus, the groundwater with slightly reduced and weak alkaline characteristics could be the result of meteoric water–wall rock interaction below the paleo–water table. Strong wall-rock buffering (from the Xiongcu Formation carbonaceous shale and slate and limestone) would have caused the oxidized and acidic fluids to progressively evolve to more reduced and alkaline conditions (cf. Rottier et al., 2016). Besides, previous S and Pb isotope studies revealed $\delta^{34}S = -3.5\text{‰}$ to $+0.1\text{‰}$, $^{206}Pb/^{204}Pb = 18.03$ – 18.43 , $^{207}Pb/^{204}Pb = 15.50$ – 15.59 , and $^{208}Pb/^{204}Pb = 37.97$ – 38.59 , which suggest a magmatic metal source, without a contribution of shallow surface strata (Xu et al., 2009; Tang et al., 2015). Therefore, we propose that the mixing of metal-rich magmatic fluid and reduced, alkaline groundwater changed the temperature, fO_2 , fS_2 , and pH conditions of the hydrothermal fluid and eventually resulted in the copper, gold, and silver ore deposition.

The final hydrothermal stage (stage C) was dominated by the actinolite + epidote + chlorite + magnetite assemblage that replaced earlier sphalerite + pyrrhotite + chalcopyrite + galena, reflecting decreasing temperature and fS_2 but increasing pH conditions in the fluid (Figs. 12G and 12J). This scenario is consistent with a system dominated by near-neutral fluids during the ore precipitation of stages A and B due to alkaline groundwater buffering, thereby progressively evolving to more alkaline conditions with minor mixing of the depleted magmatic fluid and the more alkaline groundwater toward the end of the hydrothermal process (stage C).

6.4. Genetic Significance of the Polymetallic Veins

The formation of the late polymetallic mineralization in the No. 1 deposit is thought to be the result of mixing of metal-rich magmatic fluid and alkaline, reduced groundwater via fracture networks during the late stage of porphyry mineralization. However, the genetic classification of polymetallic mineralization (Zn-Cu-Ag-Au $\pm$ Pb) in a porphyry system is controversial and a matter of considerable debate (e.g., Pudack et al., 2009; LeFort et al., 2011; Zhai et al., 2019). The polymetallic mineralization in the No. 1 deposit is characterized by sphalerite + pyrrhotite + chalcopyrite + galena and sericite + actinolite + epidote + chlorite + calcite + magnetite assemblages, lacking high-sulfidation-state sulfides (e.g., enargite, as for instance reported for the Cordilleran-type polymetallic veins; e.g., Pudack et al., 2009; Catchpole et al., 2011; Maydagán et al., 2015; Rottier et al., 2016). Moreover, Cordilleran-type veins occur in an environment similar to that of high-sulfidation epithermal deposits (Sillitoe, 2010; Kouzmanov and Pokrovski, 2012) and commonly show pipe-like ore bodies with strong upward and outward zonation, while the No. 1 deposit ore body exhibits a zonation from proximal Cu-bearing to distal Zn-Pb-bearing ores (e.g., Baumgartner et al., 2009; Rottier et al., 2016). Furthermore, the veins also lack the characteristic intermediate-sulfidation epithermal gangue mineralogy and textures (e.g., crustiform banding, symmetric banding, cockade and comb textures; Wang et al., 2019). Therefore, assigning a classification to the polymetallic veins of this study is critical to unravel their genetic evolution.

The term “subepithermal vein” has been proposed by Sillitoe (2010), and it has been used to define hydrothermal polymetallic (Zn-Cu-Pb-Ag $\pm$ Au) veins formed in fractures surrounding porphyry intrusions. These veins comprise neutral to weak alkaline pH hydrothermal miner-

als (e.g., sericite, epidote, chlorite, calcite) and contain precious and accessory mineral assemblages comparable to those observed in low- to intermediate-sulfidation epithermal deposits. In addition, LeFort et al. (2011) suggested that the subepithermal veins are transitional veins, which serve as conduits for magmatic mineralizing fluids that ascend to form epithermal deposits, and likely link the deep porphyry mineralization and the overlying low- and/or intermediate-sulfidation epithermal deposit. Consequently, the polymetallic veins in the No. 1 deposit are thought to be similar to subepithermal veins. Furthermore, detailed SEM-EDS analysis of polymetallic vein calcite revealed low Mn concentrations (<2 wt%; Fig. S8), typical for Mn-bearing carbonates (e.g., rhodochrosite, manganocalcite) in intermediate-sulfidation epithermal deposits (Wang et al., 2019). Consequently, based on their base metal (i.e., Zn-Pb) and alteration mineral assemblages, the polymetallic veins studied here constitute important transitional veins, revealing the hydrothermal evolution of porphyry to intermediate-sulfidation epithermal mineralization.

Compared with typical intermediate-sulfidation epithermal deposits (Wang et al., 2019, and references therein), the polymetallic veins have high Cu concentrations (up to 2.14 wt%; Table S2), which, however, significantly decrease with increasing distance from the center of the deposit. Furthermore, these veins display changing Cu/Au, Cu/Ag, and Au/Ag ratios with increasing distance from the center of the deposit, while low ratios occur at the distal zone (Z3) (Fig. 7). Relative to Au and Ag, the Cu concentration seems to be more controlled by the distance from the center of the deposit. Experimental studies showed that alkaline pH values result in Cu rather than in Ag and Au precipitation, but because acidic conditions prevail in high-sulfidation environments, copper solubility as chloride complexes is high (Mountain and Seward, 2003; Kouzmanov and Pokrovski, 2012) and well supported by the reactions 4a, 4b, 5, and 6. A particularly good example is provided by the skarn environment, with its commonly observed metal zonation from proximal Cu to distal Zn-Pb zones resulting from the neutralization of the ore-forming fluid by the carbonate wall rocks (Kouzmanov and Pokrovski, 2012). Additionally, thermodynamic studies suggested that the Ag solubility from bisulfide complexes would be higher than that of Au under alkaline pH conditions at medium to low temperatures (<350 °C; Gammons and Williams-Jones, 1995; Williams-Jones et al., 2009). These results support that the mixture of alkaline groundwater and Cu-, Au-, and Ag-rich magmatic fluid enables Ag and Au, compared

to Cu, to migrate even farther away from their source prior to sulfide precipitation in the veins.

Most of the intermediate-sulfidation epithermal deposits have abundant hydrothermal carbonates (Sillitoe and Hedenquist, 2003), which are interpreted to be a result of increasing fluid pH relative to moderately acidic hydrothermal fluids (Taylor, 1971). In addition, some of them are generally hosted by carbonate rocks (e.g., La Colorada Ag-Zn-Pb deposit, Mexico; Chutatas and Sack, 2004), which retain strong buffers for ore-forming fluids. Our study indicates that the original magmatic ore-forming fluids for an intermediate-sulfidation epithermal deposit may have been Cu enriched, but premature precipitation of Cu is thought to have occurred due to mixing with alkaline groundwater.

In conclusion, the model proposed for the polymetallic veins emphasizes a genetic relationship between porphyry mineralization and intermediate-sulfidation epithermal deposits and explains why the intermediate-sulfidation epithermal deposits lack copper compared to high-sulfidation epithermal deposits in porphyry copper systems. We expect that the discovery of this relationship will provide an important guide for regional Cu exploration in porphyry copper systems.

7. CONCLUSIONS

(1) The polymetallic mineralization is the youngest style of mineralization in the No. 1 deposit and overprints early porphyry-type mineralization. It is divided into three stages, from early to late: pyrite + sericite (stage A), sphalerite + pyrrhotite + chalcopyrite + galena + quartz (stage B), and actinolite + epidote + chlorite + magnetite + quartz (stage C). The stage B assemblage represents the main ore mineralization stage.

(2) A decrease of the oxidation state and fS_2 of the fluid occurred from stage A to stage B, followed by a progressive decrease of the fS_2 and a pH increase of the fluid during stage C, accompanied by decreasing temperatures.

(3) The integrated O isotopic compositions suggest that the ore-forming fluid was a mixing product of magmatic fluid and groundwater.

(4) Decreasing temperature, $\log fO_2$, and $\log fS_2$ and increasing pH conditions of the hydrothermal fluid, attributed to the mixing of metal-rich magmatic fluid separated from magma with slightly reduced and alkaline groundwater, resulted in copper, gold, and silver ore deposition.

(5) The polymetallic veins may have acted as a conduit between porphyry and intermediate-sulfidation epithermal mineralization, while the ore-forming fluid derived from mineralized

magma is thought to have lost some of its metal (e.g., Cu) endowment due to deep-seated interaction with alkaline groundwater prior to reaching an epithermal environment, where relatively Cu-poor intermediate-sulfidation epithermal deposits formed.

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